UTILIZATION OVER QUBITS: RYDBERG REGISTERS IN PHYSICS WORLD MODELS

Anonymous authors
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ABSTRACT

In hybrid classical–quantum models, how many qubits are available matters less than how well they are used. We attach an input-dependent Rydberg-atom co-processor, QUOTA, to world models on ten scientific and optimization tasks (fusion, geothermal, fluid flow and vehicle routing; released checkpoints where available), under decision rules registered before each run, most after earlier test results on the same rows. With the same 30 qubits, routing cuts error by up to 36%, against 7% for the standard design that averages features before the qubits see them; a linear corrector on the same routes, writing spatial modes that averaging cannot reach, already gets 29 of the 36 points, input-adaptive routing adds 3–4 points, and a similarly sized classical core behind the same routing design gains as much (at 100 qubits, ten 10-atom groups, on shear flow the register beat a smaller graph-blind MLP, 199 against 244 parameters, on five new seeds by 2.5%, but a registered follow-up did not establish that edge against a graph-aware GNN of similar size or a wider MLP). With the backbone fixed (the rest retrained), error falls as the budget grows to 100 qubits, by 16% on shear flow and 8% on convection, and levels off at 200 although the captured share still rises; registered bounds for 50, 100 and 200 qubits (oracle-coverage diagnostics from test labels, not forecasts) held on every seed. The gains come from allocating by signal type: a router trained on residuals predicts, from the input and the backbone's forecast, which frequency modes (Fourier octaves and wavelet tiles) to correct; selected modes get a classical bypass plus the register, the rest a classical head. Atom positions, detunings and tethers between atom groups are also set per input, but in a registered control the tested geometry and tethers did not pass the rule for a gain. At the sizes we can simulate, we detect no consistent quantum superiority over matched classical cores. A descriptive closed form, $G(K) = \eta U_\infty[1 - (1 + K/K_0)^{1-a}]$ with $a \approx 2$, tracks the share of error energy removed by K qubits, where U_∞ is the output basis's reachable share (for any core) and η a core's fill.

1 INTRODUCTION

Learned physics simulators are now central to scientific machine learning, from turbulence and magnetohydrodynamics (Ohana et al., 2024; McCabe et al., 2025) to tokamak plasmas (Boschi et al., 2026; Rousseau et al., 2026) and real-world flow measurements (Hu et al., 2026). In parallel, a fast-growing literature inserts small quantum circuits into such models: hybrid neural operators (Jain & Kalev, 2026; Hossain et al., 2026; Marcandelli et al., 2025), quantum adapters for frozen networks (Koike-Akino et al., 2025; Liu et al., 2025; Raj & Coyle, 2025), and quantum-informed priors for turbulence (Wang et al., 2026a; Xue et al., 2026). Whether the quantum part does what a classical module with the same interface and parameter count could not is rarely tested; where tested, apparent edges vanish under equal-tuning classical baselines (Bowles et al., 2024; Pandey, 2026a).

We run that test at scale, on the analog neutral-atom model family that current hardware natively evolves (Kornjača et al., 2024). QUOTA (*Qubit Utilization thrOugh Trained Allocation*) is a Rydberg register (one atom is one qubit) whose allocation is learned, not hand-placed. Per-atom detunings are functions of each input; geometry and drives are trained by gradient; register size and graph are chosen by architecture search; band-limited or routed insertions decide which modes (or, per input, which error tiles) the atoms serve. It is inserted into the *released* checkpoints of pub-

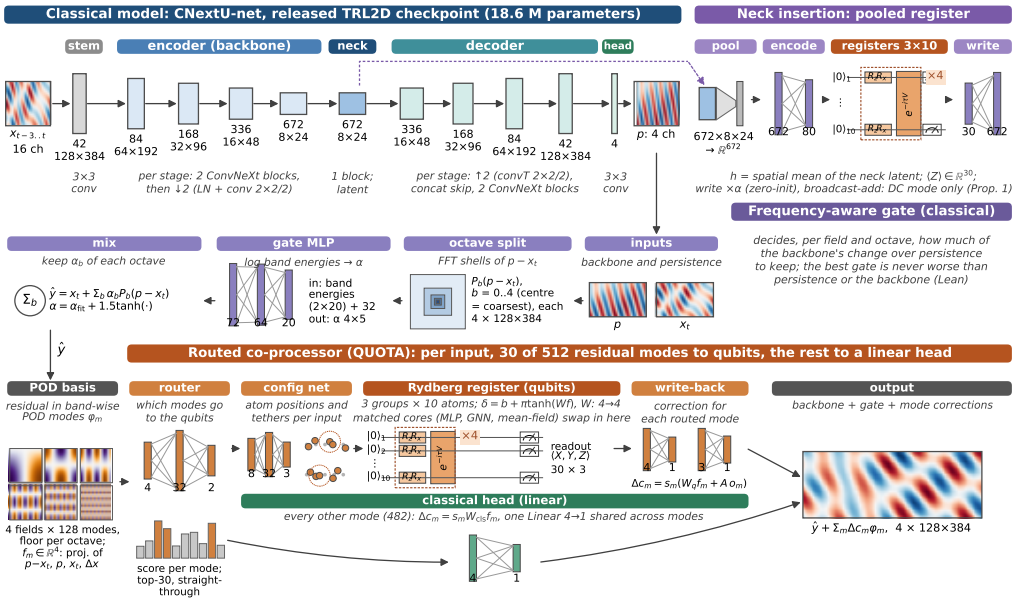


Figure 1: Architecture with tensor sizes (TRL2D). Each module has a coloured band (its name) and a line on its role. Top: the released CNextU-net (slabs: channels, height×width) and the standard pooled register in its neck (Prop. 1). Middle: the frequency-aware gate. Bottom: the routed co-processor, with the register at gate level; matched classical cores swap into its slot. Bars: fully connected layers. Other backbones: Appendix B.

lished models and compared with a parameter-matched MLP core behind the identical interface, a no-interaction ablation, and a mean-field control, under kill rules fixed before each run. The premise is a division of labor: a pretrained classical model handles the easily digitized part of a field, and a small qubit budget is worth only as much as it is utilized on the part each input gets wrong. We separate two claims. Utilization, not qubit count, governs the gain. Whether a quantum core exploits a given utilization better than a classical one is a second claim; at the sizes we can simulate we detect no consistent quantum superiority, and we claim no equivalence margin.

Contributions.

1. **Input-dependent allocation of qubits** (§3, §5.5, Fig. 1): each prediction's error is split into Fourier octaves and band-wise POD modes; a router picks input-dependent modes, mostly coarse, for a classical bypass plus 30 Rydberg atoms whose positions, detunings and tethers are set per input, the rest to a classical head, behind a frequency-aware gate between backbone and persistence. With the same 30 atoms error falls by 0.9–36% across rows, 36% on convection against 7% pooled; a linear corrector writing non-DC modes gives 29 of the 36 points, adaptive routing 3–4, and matched classical cores gain as much.
2. **Hybrid scaling with the qubit budget** (§5.5, Appendix P): with the classical model fixed, error falls from 0 to 100 qubits, by 16% on shear flow and 8% on convection, and levels off at 200, tracked by a descriptive closed form; bounds for 50, 100 and 200 qubits, registered before the runs, held on all 18 seeds.
3. **Proofs of where qubits can help** (§5.3, Lean 4): a register fed spatially averaged (*pooled*) features, the standard design, reads and writes only the DC mode of the latent it is attached to (Proposition 1); routed corrections are bounded per Fourier band; the optimal band gate is never worse than persistence or the backbone.
4. **A matched-control audit on released models** (§5.1): eight physical benchmarks (turbulence, MHD, convection, shear flow, tokamak transport and equilibrium, geothermal flow), real-world flows (RealPDEBench) and vehicle routing with two released solvers (Appendix R), with similarly sized classical cores (MLP, GNN on its own learned graph,

mean-field register) in the register's slot; a long-horizon TRL2D edge (§5.2) did not replicate on a registered fourth seed.

2 RELATED WORK

Quantum modules in classical surrogates. Hybrid FNO and U-Net variants (Jain & Kalev, 2026; Hossain et al., 2026; Marcandelli et al., 2025) and quantum-informed turbulence priors (Wang et al., 2026a; Xue et al., 2026) typically report gains without parameter-matched cores or released-checkpoint anchors; the closest in-model study, a 5-qubit mixer in a 3-D FNO, beat a larger dense mixer but not its matched MLP-mixer (Papierz et al., 2026). Quantum adapters for frozen language models (Koike-Akino et al., 2025; Liu et al., 2025) use quantum circuits to *generate* classical weights, so inference is classical by construction; our inference-inertness finding reaches the same deployment regime empirically, the classical-surrogate regime of Schreiber et al. (2023). Reservoir and attribution studies show that apparent quantum edges shrink under equal-tuning classical baselines (Pandey, 2026a;b) and reach nulls with matched controls and multiplicity correction (Chen, 2026; Gillani et al., 2026; Legziel Levy et al., 2026; Herbort et al., 2026); the register-free QSWM finds a short-horizon edge on cellular automata (Jiang et al., 2026). We follow the benchmarking cautions of Bowles et al. (2024) and Schuld & Killoran (2022), adding pre-registration, kill rules, and released-checkpoint anchors across eight domains.

Analog ansätze, dequantization, and scaling. Rydberg reservoirs (Bravo et al., 2022; Kornjača et al., 2024) and quantum neural operators (Wang et al., 2025) report gains against quantum or unmatched baselines. ReCon co-trains atom positions and drives derivative-free (DiBrita et al., 2024), ResQ feeds inputs into detunings on fixed lattices (DiBrita et al., 2025), and PulserDiff differentiates pulses and positions (Abramavicius et al., 2025); position gradients were argued ineffective under the divergent r^{-6} term (de Keijzer et al., 2025), which we bound by clamping distances while training geometry by exact statevector autodiff. Variational models that are both trainable and non-dequantizable exist, but on specially constructed learning problems (Gil-Fuster et al., 2025), and trained quantum models admit classical shadows (Keller et al., 2026); our dequantization suite measures this on a trained world model. Saturation in qubit count has been reported in non-parameter-matched sweeps (Kim et al., 2025; Vyskubov et al., 2026; Zabihi et al., 2026) and is predicted by a frozen-block bound (Qi et al., 2026); our ladder adds a 0-qubit arm and a matched core.

Spectral limits of neural surrogates. Neural networks fit low frequencies first (Rahaman et al., 2019), and neural PDE solvers lose long rollouts through high-frequency error (Lippe et al., 2023). We give the mirror-image statement for insertion modules: a pooled module reads and writes only the DC mode, whatever its core (Proposition 1).

3 QUOTA: THE REGISTER, ITS INSERTION, AND THE CONTROLS

Insertion. A backbone produces a latent field $z \in \mathbb{R}^{d\times N}$ over N spatial positions. The standard insertion pools $h = \frac{1}{N}\sum_s z_{:,s}$, applies a core $c(h) \in \mathbb{R}^m$, and writes back a zero-initialized gated broadcast $z_{:,s} \leftarrow z_{:,s} + \alpha \odot U c(h)$ for every s ($\alpha = 0$ at initialization, so the inserted model starts exactly at the released checkpoint). Six audit rows insert at a field-valued latent (a U-Net or operator neck); two (the Sophelio starter model and TokaMind) insert at a vector latent, where the core sees the whole latent and no pooling occurs. §5.4 replaces this insertion by band-limited and routed ones.

Register core. Each 10-atom register evolves under $H/\hbar = \sum_i \frac{\Omega_\ell}{2} X_i - \sum_i \delta_{\ell,i} n_i + \sum_{i<j} C_6 |r_i - r_j|^{-6} n_i n_j$ with input-conditioned detunings $\delta_{\ell,i} = b_{\ell,i} + \pi \tanh(W_\ell h)_i$ (W zero-initialized in the pooled neck, Gaussian with s.d. 0.1 in the routed co-processor), trainable drives Ω_ℓ, times τ_ℓ and 2-D atom coordinates r_i (distances clamped at $0.5\,R_b$), L=4 layers of a first-order split (per-atom $R_z(\delta)R_x(\Omega)$, then the diagonal interaction phase), and $\langle Z_i\rangle$ readout (the routed co-processor also reads $\langle X_i\rangle, \langle Y_i\rangle$). Thirty atoms are three blockade-decoupled registers, keeping trainability fixed across the qubit ladder (Cerezo et al., 2025). Training uses an exact complex64 statevector propagator with autograd, cross-checked against CUDA-Q ($\max_i |\Delta\langle Z_i\rangle| \le 10^{-6}$, 12 qubits).

Controls. Every control implements the identical pool→core→residual interface at matched total parameters (the MLP core's width is set from the register's allocated parameter count; at the TRL2D neck the cores differ by 124 parameters, 0.08% of the core and under 0.001% of the 18.6-million-parameter model). *Matched MLP:* a one-hidden-layer MLP core. *No-interaction:* $V_{ij} \equiv 0$ from step 0 (a product state; removes entanglement and the interaction nonlinearity). *Mean-field (Gutzwiller):* the interaction phase replaced by its Hartree decoupling $\theta_i = \tau \sum_{j \neq i} V_{ij} \langle n_j \rangle$, which keeps the $V_{ij}(r)$ nonlinearity and trainable geometry but forbids entanglement. The ladder removes structure, then interactions, then entanglement alone.

Cost and deployment. Every experiment, from fine-tuning the released backbones to exact register simulation, ran on a single NVIDIA DGX Spark (one GB10 chip, 128 GB unified memory). Exact simulation is the price: one TRL2D fine-tuning step (released CNextU-net, forward, backward and AdamW, batch 4) takes 504 ms for the backbone, 511 ms ($+1.4\%$) with the MLP core, and 595 ms ($+18\%$) with the 30-atom register. Trained neck registers are inference-inert, so deploying them needs no quantum hardware (Appendix O).

Probabilistic variant. Following functional generative networks (Alet et al., 2025), a noise draw ξ_m enters the configuration network, so each draw gives the input its own atom positions, detunings and tethers and one ensemble member $\hat{y}_m$. Training minimizes the fair energy score (Gneiting & Raftery, 2007), $\mathrm{ES} = \frac{1}{M} \sum_m \|\hat{y}_m - y\| - \frac{1}{M(M-1)} \sum_{m<m'} \|\hat{y}_m - \hat{y}_{m'}\|$ (M=2, norms in VRMSE units, i.e. divided by each window's own target variance, which makes it observation-weighted and not strictly proper), and a validation-tuned scale τ sets the spread (Appendix Q).

4 Protocol

Hypothesis and kill rule. H1: the register improves the benchmark metric over a parameter-matched classical core trained with identical hyperparameters and step budget; if the core is statistically indistinguishable or better across seeds, H1 is rejected for that regime (rule fixed before each round). Each benchmark has one primary row fixed a priori. Tuning (Optuna, train/validation only) gives every model, backbone included, the same trial budget; TEST is scored once.

Statistics. Per row, seed-paired relative gains, paired t tests and the paired 95% CI half-width; across rows, a tie-aware sign test and a random-effects pool (Table 4); on TRL2D, per-trajectory Wilcoxon tests, Holm-corrected. $\pm$ is the standard deviation over three seeds.

Pre-registration. Hypotheses, kill rules and decision rules were timestamped before the corresponding runs (self-attested for review; code, logs and the record are released); the one extension registered after its models' results were known (§5.2) is labelled as such. Each final configuration saw TEST once, but some later protocols were designed after earlier TEST results on the same rows; Appendix H labels every claim confirmatory or exploratory accordingly.

5 Results

5.1 The register ties a matched classical core across eight benchmarks

Table 1 reads left to right: published state of the art, our classical model, and the same model with a matched MLP or with qubits. The rows use the released checkpoints of The Well's CNextU-net (Ohana et al., 2024) and of TokaMind (Boschi et al., 2026) on TokaMark (Rousseau et al., 2026), the official U-Net pipelines of a groundwater heat-transport surrogate (Pelzer et al., 2025) and of the Fusion Equilibrium Challenge (Waller et al., 2026), a TORAX benchmark (Citrin et al., 2024), and our FNO arena. Against the best published numbers (PDE-FM (Soares et al., 2025), PhysiX (Nguyen et al., 2025), Flowers (Muser et al., 2026), EddyFormer (Du & Krishnapriyan, 2025)), our shear-flow hybrid is lower (0.0207 vs 0.0345, full test); a gated three-backbone ensemble is numerically close on convection (0.0403 vs 0.0415; best single model 0.0425) and 1% lower on TRL2D (0.1909 vs 0.1930, same 4→1 protocol), with more compute; on MHD a retrained Flowers backbone with a metric-aligned fine-tune is 1.0% above as one model (0.1172 vs 0.1160) and at parity as a two-backbone ensemble (0.1154); TokaMark is matched by one Scratch run (0.1379 vs

Table 1: From the published state of the art to classical + qubits (test split; lower is better except S; three seeds unless noted). Published: PDE-FM (shear flow, convection), Flowers 4→1 (TRL2D), EddyFormer (MHD), CNextU-net (rollout), released U-Net (RealPDEBench), RouteFinder, RADAR (RRNCO), CaliciBoost (Caco-2; ¶not re-derivable from training data, App. E); #+test-time search). Our best: strongest classical result (Appendix G; ^{e}ensemble), bold if below published; other bold: best register-comparison arm; vs columns at 30 qubits (seeds better in parentheses). Top: routed co-processor on a fixed fine-tuned CNextU-net; persistence 0.0646, 0.2024, 0.5761, 0.418; convection on every test window. Middle: pooled register in each neck (†two length-matched seeds; ‡backbone tuned under an earlier objective). Bottom: register in the model's own latent (rollout four seeds, RouteFinder and Caco-2 five; §bf16). Full statistics: Table 4.

Benchmark	Published	Our best	Our classical	+MLP	+30 qubits	+50	+100	vs MLP	vs classical
Routed co-processor: input-dependent qubit configuration									
shear flow	0.0345	**0.0198**	0.0236	0.0213	0.0207	0.0205	**0.0198**	+2.6%	+12.3%
convection (RB)	0.0415	**0.0403**^{e}	0.0456	0.0435	0.0430	0.0425	**0.0418**	+1.1%	+5.6%
turbulence (TRL2D)	0.1930	**0.1909**^{e}	**0.1922**	**0.1922**	**0.1922**	**0.1922**	**0.1922**	0.0%	0.0%
MHD (MHD_64)	0.1160	**0.1154**^{e}	**0.1733**	**0.1733**	**0.1733**	**0.1733**	**0.1733**	0.0%	0.0%
Standard design: pooled register in the neck									
FNO arena (TRL2D data)	—	—	0.9567	0.9534	**0.8447**	—	—	+11.4% (3/3)	+11.7% (2/3)
MHD (MHD_64)†	0.1160	**0.1154**^{e}	0.1897	0.1897	**0.1896**	—	—	+0.0% (2/2)	+0.0% (2/2)
convection (RB)	0.0415	**0.0403**^{e}	0.4563	0.5275	**0.4221**	—	—	+20.0% (3/3)	+7.5% (1/3)
shear flow	0.0345	**0.0198**	0.2686	**0.2539**	0.2659	—	—	−4.7% (2/3)	+1.0% (2/3)
tokamak transport (TORAX)	—	—	0.0158	**0.0141**	0.0168	—	—	−19.2% (0/3)	−6.4% (2/3)
geothermal plume (1HP)‡	—	—	0.0291	0.0292	**0.0288**	—	—	+1.3% (2/3)	+1.2% (2/3)
fusion equilibrium ($S\uparrow$)‡	0.9959	0.9949	0.7905	0.7848	**0.7929**	—	—	+1.0% (2/3)	+0.3% (1/3)
tokamak signals (TokaMark)	0.140	**0.1334**^{e}	0.1509	0.1504	**0.1501**	—	—	+0.2% (1/3)	+0.5% (2/3)
Neck rows better / worse / tied (sign test)								4 / 2 / 2 (p=0.69)	5 / 1 / 2 (p=0.22)
turbulence, rollout 13:30 (TRL2D)	1.01	**0.9892**	0.9892	1.0101	**0.9660**	—	—	+4.4% (2/4)	+2.3% (1/4)
real flow, foil (RealPDEBench)	0.00948	**0.00931**	0.00948	**0.00939**	**0.00939**	—	—	0.0%	+1.0%
routing, RouteFinder (gap %)§	8.58	**6.02**#	8.65	—	**8.61**	—	—	—	+0.5% (2/5)
routing, RRNCO ACVRP (gap %)	2.61	**2.41**#	**3.49**	—	3.71	—	—	—	−6.2% (0/3)
molecules, Caco-2 (TDC, MAE)	0.256¶	0.276	0.362	**0.345**	0.437	—	—	−26.6%	−20.7%

0.140) and 4.7% lower with a post hoc three-run ensemble (0.1334). The margins come from the classical part, a frequency gate with persistence; the routed co-processor adds 12–13% on shear flow and 5–6% on convection, where the error is predictable (Figure 2), and matched classical cores match it. In the neck no comparison is significant after correction (sign test $p = 0.69$ against the matched core; Figure 6). Separately, a pooled register reads and writes only the spatial mean of its latent (Proposition 1); we do not test whether this explains the null. Where a fine-tuned model beats a published number, the backbone alone beats it within seed noise, so those gains belong to fine-tuning (Appendix G); four neck rows are imprecise (paired 95% CI half-widths 53–99%, Appendix A).

5.2 A CANDIDATE EFFECT THAT DID NOT REPLICATE: LONG-HORIZON ROLLOUTS ON TRL2D

On TRL2D we fine-tuned the official CNextU-net in two registered rounds. With a one-step objective every model improves on the checkpoint indistinguishably (backbone 0.19307 ± 0.00008, MLP 0.19314 ± 0.00019, register 0.19315 ± 0.00009). Trained through rollouts (Brandstetter et al., 2022) and scored on The Well's long-horizon window 13:30 (published 1.01), the register was the only model below the checkpoint on three seeds ($\mathbf{0.9528 \pm 0.0542}$ vs 1.0301 ± 0.0449 and MLP 1.0322 ± 0.0259), but a registered fourth seed did not replicate the edge (Table 11, Appendix J).

5.3 WHAT POOLING CONSTRAINS: THE INSERTION'S DC LATENT MODE

Proposition 1 (a pooled register sees only the spatial mean of its latent)**.** *Let $P(z) = \frac{1}{N}\sum_{s=1}^{N} z_s$ be mean pooling over N positions and let g satisfy $\sum_s g_s = 0$. Then $P(z+g) = P(z)$; in particular P is invariant to adding any multiple of a non-DC discrete Fourier mode ($\zeta^N = 1$, $\zeta \neq 1$). A spatially constant write-back c is orthogonal to every zero-mean field, $\sum_s c\, g_s = 0$. Hence the insertion of §3 reads and writes only its latent's DC spatial mode, whatever its core. In any orthonormal basis, a perfect correction of the coefficients in a set T leaves exactly the energy outside T.*

All four statements are proved in Lean 4 with Mathlib (de Moura & Ullrich, 2021; The mathlib Community, 2020), standard axioms only (Appendix C).

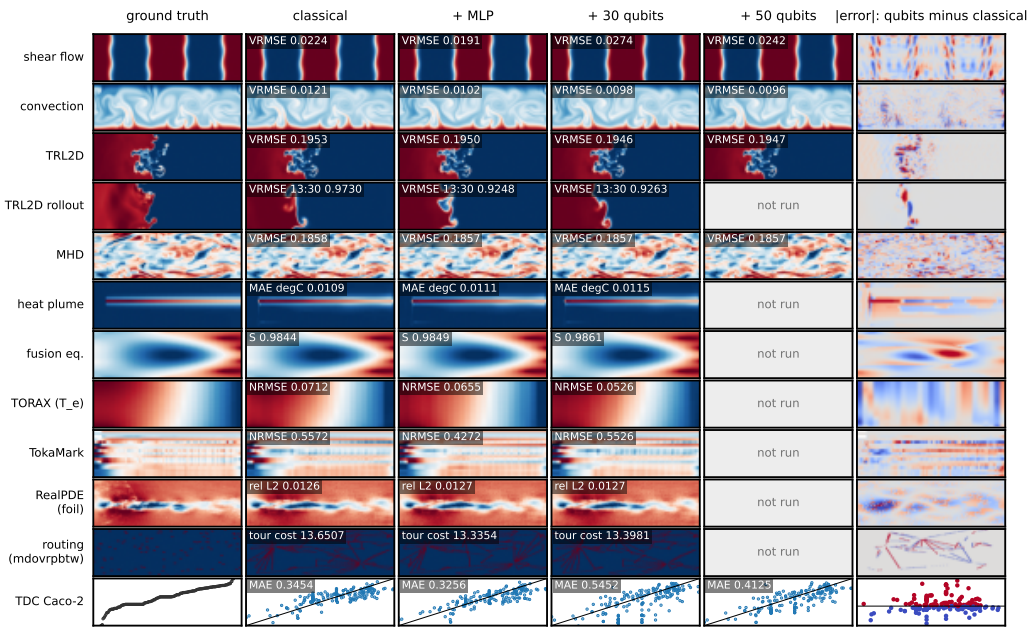


Figure 2: Model outputs against ground truth, one sample each, fixed in advance, on every benchmark with register arms (not RRNCO). Columns: classical, + matched MLP, +30 and +50 qubits (routed co-processor of Table 1, top, on shear flow, convection, TRL2D and MHD; TRL2D rollout: step 22 of 30; the pooled neck register elsewhere; "not run": no 50-qubit model), and the change in absolute error from classical to the largest register shown (blue: lower). Panels carry that sample's error (S for fusion: higher is better). TDC Caco-2: predicted against true permeability; routing: tours. Notes per row: Figure 17.

Table 2: Correcting only the low-frequency or high-order part of a model's prediction (test split unless noted; three seeds). RealPDEBench airfoil: relative L_2 on 5,140 test windows; the corrector acts on the lowest $3\times4\times4$ space–time Fourier modes of the prediction. Fusion equilibrium (Sophelio): score S on validation; the corrector writes PCA components $k \geq 8$ of a frozen model.

Model (benchmark)	Model alone	+linear	+MLP	+qubits
U-Net, released (RealPDEBench) ↓	0.00948	**0.00937**	0.00939	0.00939
FNO, released (RealPDEBench) ↓	0.01413	0.01292	**0.01267**	**0.01267**
frozen NAS model (Sophelio, validation) ↑	0.9864	—	0.9876	0.9879

Call the share of the error energy in the modes an insertion can read and write its *utilization ceiling*. The proposition concerns the insertion latent; a downstream decoder could spread a DC latent change over spatial outputs, so for pooled insertions we report the *measured* DC share of the output error, not a proved ceiling. On TRL2D (Table 5, Appendix C) the DC shell holds **2.9%** of the fine-tuned model's one-step error energy; the backbone misses 8–11% of the one-step change at $1 \leq |k| \leq 4$ but 34–45% at $|k| \geq 17$, the spectral bias of neural surrogates (Rahaman et al., 2019; Lippe et al., 2023). Paired over seeds, the register's only measurable effect is at k=0 ($-0.49\% \pm 0.21$ of that band's error, 3/3 seeds; matched MLP $+0.10\% \pm 0.19$); every other band moves within noise. The classical core is confined to the same mode, which is why the two tie.

5.4 Raising utilization: aiming the insertion where the error lives

If pooling is capped by the DC share, an insertion aimed where the error lives should help, and help any core (Table 2). On the RealPDEBench airfoil (Hu et al., 2026), a low-mode corrector lowers the released U-Net's test relative L_2 from 0.00948 to 0.00937 and the FNO's by 10%; a linear map captures most of it, and the register equals its matched MLP to four digits. On the Fusion

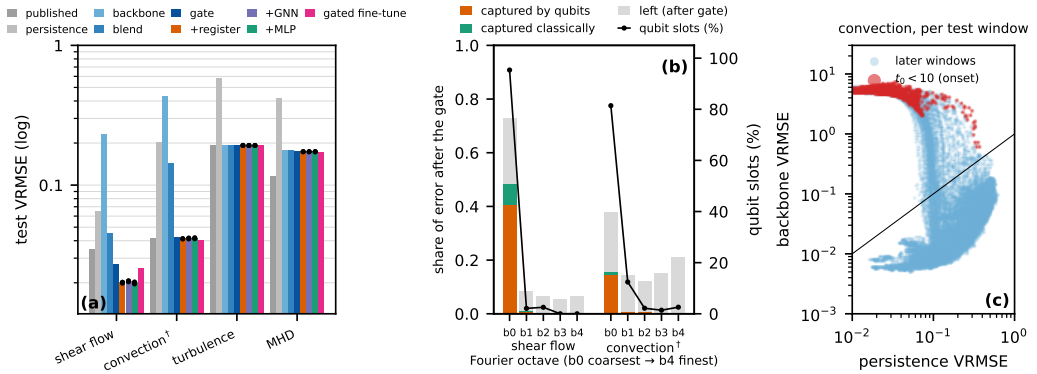


Figure 3: Where the error sits on the four Well benchmarks (shear flow, Rayleigh–Bénard convection, turbulent radiative layer, 3-D MHD). (a) Test VRMSE, log scale, from the published best model through persistence ($\hat{y}_{t+1}{=}x_t$), our backbone, the frequency gate, and the gate plus qubits, GNN or MLP (dots: seeds; †stratified test sample). (b) After the gate, the share of the remaining error in each Fourier octave captured by the qubits, by the classical head, or left; the line gives the router's share of the 30 qubits per octave: 80–95% go to the coarse octave that holds the predictable error. (c) Convection, per test window: at convection onset (red) the backbone is 10–100× worse than persistence; the gate corrects it.

Equilibrium Challenge (full-data pipeline), a coupling that writes only the fine PCA components lifts VALID S by $+0.0012$ to $+0.0015$ with either core; the register leads its matched MLP by $+0.0002$ ($p \geq 0.41$), or by $+0.0010$ ($p = 0.09$ corrected) with a register-selected learning rate (Appendix D).

Evaluation error, persistence, and a frequency-aware gate. On rayleigh_benard and shear_flow, persistence beats The Well's published baselines and every backbone we fine-tuned, though not the foundation models PDE-FM (both rows) and PhysiX (convection) (Fig. 3a): the benchmark's per-window VRMSE is dominated by low-variance windows that MSE-trained backbones barely weight. A causal gate between backbone and persistence per Fourier octave (its optimum is proved never worse than either) cuts shear_flow's test VRMSE from 0.2297 to 0.0272. Routing then sends the reachable error to the qubits (Fig. 3b), and the register, a GNN on the same graph and an MLP all reach about 0.020 (Appendix I). Trained end to end with the metric as loss, a three-backbone gated ensemble reaches 0.0403 on rayleigh_benard (full test; persistence 0.2024).

5.5 Which input-dependent allocation? A registered cross-benchmark study

Design. To separate *where* the atoms work from *what* computes there, we freeze each backbone and train a co-processor to correct its residual $r = y - p$ in an orthonormal multi-level Haar basis (r is the training target; at inference the co-processor sees only causal inputs and p). Per input, a strategy picks the K tiles the atoms read and write: *pooled* (the audit's insertion), *static*, *proxy*, *learned*, *gain* (the budgeted optimum) or *random* (Appendix E). Atoms sit at their tiles, so the allocation sets the interaction graph, and every core starts as the same linear corrector. Proved bounds fix what can be measured: the captured share never exceeds the *utilization ceiling* U, ceilings nest, and a broadcast correction removes at most the DC energy. Rows: TRL2D, rayleigh_benard, shear_flow and MHD_64 with the released CNextU-net, RealPDEBench with its released U-Net and FNO, and the heat plume with the 1HP_NN U-Net; $K = 30$, three seeds, every choice on validation, test once; the protocol and its two amendments were registered before the runs.

Allocation sets the ceiling; predictability sets how much of it is reached. With the same 30 atoms, routing raises the ceiling far above the pooled DC share (Table 3; e.g. from 0.039 to 0.34 on shear_flow and from 0.047 to 0.46 on the heat plume), and where the backbone's error is systematic the gain follows: routed atoms cut mean VRMSE by 36% on rayleigh_benard and 26% on shear_flow against about 7% pooled (both relative to released backbones that persistence beats in the mean, Appendix I; a linear corrector alone: 29% and 26%, so the cores add 7.5 points and none);

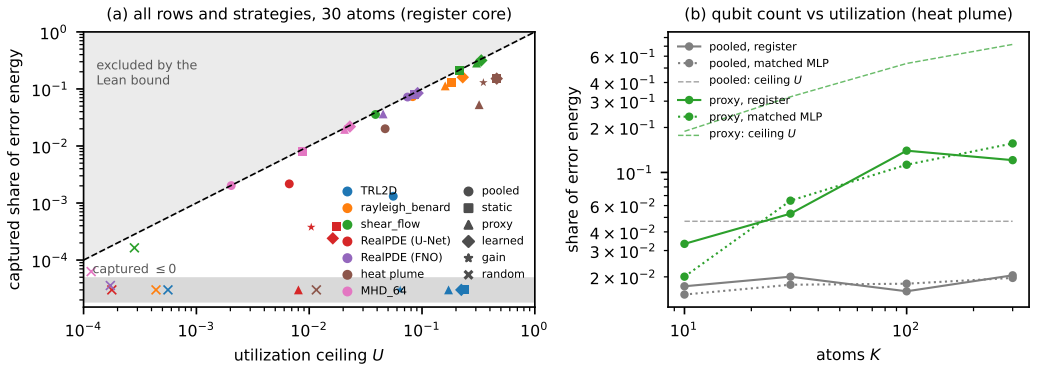


Figure 4: Allocation study (frozen backbones, test split, three seeds). (a) Captured share of each backbone's error energy against the utilization ceiling U of the allocation, for every row and strategy with the register core; no point can lie above the diagonal. (b) On the heat plume, adding atoms to the pooled insertion changes nothing (its ceiling is the DC share); routed atoms keep improving as their ceiling grows. Cores coincide throughout (dotted: matched MLP).

Table 3: Allocation study, 30 atoms, register core, test split, mean of three seeds. Pooled: the audit's insertion (ceiling = DC share). Best: the register's validation-best tile strategy, its utilization ceiling U, the captured share of error energy and the test-metric gain over the frozen backbone (VRMSE; relative L_2 on RealPDEBench); the last column is the register's seed-paired gain over its matched MLP at that strategy. On TRL2D and RealPDEBench (U-Net) every tile strategy captures at most 0.0004, and the listed one is the tie-broken validation choice. rayleigh_benard's mean gain is tail-driven: the convection onset (4% of windows) gives 47%; median VRMSE -0.3 to -0.5%. †Released checkpoint does not reproduce its published number under our evaluator (Appendix G).

	pooled		best tile strategy				
Row (frozen backbone)	DC share	gain	strategy	U	captured	gain	reg. vs MLP
TRL2D (CNextU-net)	0.056	+0.06%	static	0.238	0.000	+0.00%	+0.00% (0/3)
rayleigh_benard (CNextU-net)	0.083	+6.69%	proxy	0.161	0.113	+36.25%	−0.48% (1/3)
shear_flow (CNextU-net)†	0.039	+6.77%	learned	0.335	0.316	+26.29%	+0.03% (3/3)
MHD_64 (CNextU-net)†	0.002	+0.11%	gain	0.023	0.022	+0.87%	+0.00% (1/3)
RealPDEBench (U-Net)	0.007	+0.12%	static	0.017	0.000	+0.01%	−0.00% (0/3)
RealPDEBench (FNO)	0.074	+3.32%	gain	0.091	0.085	+3.87%	+0.01% (3/3)
Heat plume (1HP_NN U-Net)	0.047	+1.45%	static	0.459	0.152	+9.23%	−0.18% (1/3)

on the heat plume 30 static atoms lower error by 9.2% against 1.45% pooled (oracle 15%), and the matched MLP does as well; RealPDEBench's FNO reaches 93% of U. On TRL2D nothing is captured by any strategy or core, even under oracle placement with a 44% ceiling. Yet with the residual supplied as a feature the same pipeline captures 99.7% of it, so the released turbulence backbone's fine-scale error is not predictable by these correctors from causal inputs. The registered predictions split accordingly (captured never exceeded U; captured rises with U across the 42 row–strategy cells, Spearman $\rho = 0.71$ with row-bootstrap 95% CI $[0.18, 0.98]$; but routing beats pooling on 5 of 7 rows, failing exactly on the two where no tested corrector captured the error, and gain-predicted routing beats energy routing on only 3 of 7, within noise of learned and proxy routing).

Qubit count versus utilization. Figure 4b shows this on the heat plume: 10 to 300 pooled atoms capture 1.5–2.1% (ceiling 4.7%), while routed atoms capture 2, 6, 11 and 16% as their ceiling grows from 19% to 72%. With the backbone fixed (the rest retrained per budget), the hybrid's error falls with the budget, by 12, 13 and 16% on shear_flow and 5, 6 and 8% on rayleigh_benard at 30, 50 and 100 qubits, and levels off at 200 (15% and 8%; on shear flow the change from 100 to 200 is not resolved with three seeds, paired $p = 0.41$, while the captured share still rises on every seed) (MLP at 100: 11% and 8%). A descriptive closed form, $G(K) = \eta\, U_\infty[1-(1+K/K_0)^{1-a}]$, tracks this: the reachable share $U(K)$ is an oracle coverage computed from test labels (so it describes, and

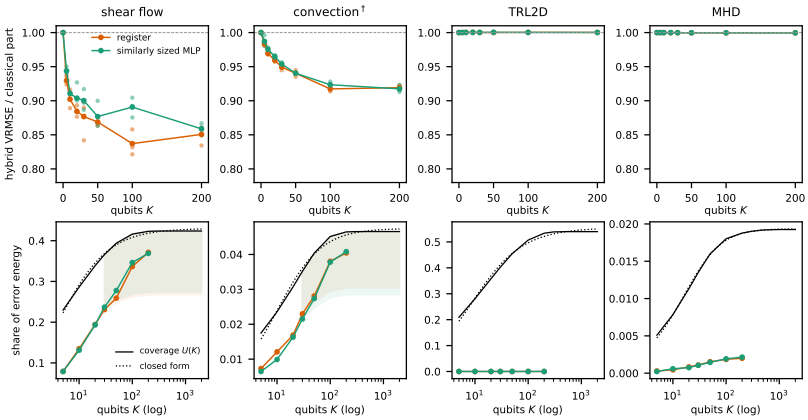


Figure 5: Hybrid scaling with the qubit budget, backbone fixed, rest retrained per budget (three seeds; dots are seeds; K counts atoms in groups of at most 10, so the largest joint quantum state has 10 atoms). Top: test VRMSE relative to the classical part (K=0). Error falls with each added atom group up to 100 qubits on shear flow and convection (†stratified test sample), where the correctors capture the backbone's error, and levels off at 200 while the captured share still rises; it stays at 1 on TRL2D and MHD, where they capture almost none of it. Bottom: captured share of the pre-gate anchor's error energy (register, MLP) against the reachable share $U(K)$ (solid) and the closed form of $G(K)$ (dotted); the 50-, 100- and 200-qubit points fell inside ranges registered before they ran.

does not forecast) and follows the closed form within 0.015 (a 1.85–2.62); the registered bounds $[\eta_{\text{prev}}U(K), U(K)]$ for 50, 100 and 200 qubits held on all 18 seeds (Figure 5, Appendix P), and a prospective test on unseen rows, predicted from validation data only, is registered. In a registered control (same routing design, each model trained separately) a fixed layout without tethers matches per-input positions and tethers within 0.1%, and a 3-D layout does not help (Appendix P).

The core. At each row's validation-best strategy the register ties its matched MLP (3 rows better, 3 worse, 1 tied; random-effects $+0.006\%$ $[-0.005, +0.017]$), as it does the mean-field register and a 4× MLP; a register placed beside an MLP on the same atoms ties two MLPs (-0.43% to $+0.01\%$ on the five rows with predictable error, none significant). A controlled probe finds no analog edge, and on targets generated by interacting Rydberg dynamics the register beats matched MLPs and GNNs but not its classical mean-field twin (Appendix E). A probabilistic variant is in Appendix Q: under a registered proper score both the register and its matched MLP beat the deterministic hybrid and a tuned dressing on every seed.

6 Limitations and Conclusion

Most primary comparisons use three seeds (exceptions are marked in the tables and appendix), four audit rows have paired 95% CI half-widths of 53–99%, and TEST sets were reused across protocol amendments (Appendix H). Registers are simulated exactly in groups of at most ten atoms (Liu et al., 2025); decoherence is unmodelled (readout and calibration tests: Appendix O). The allocation study freezes backbones and evaluates test samples on four rows.

A hybrid model should budget utilization, not qubits: atoms aimed at the error a backbone gets wrong predictably turn a fixed budget into gains that grow with budget, whereas a pooled insertion reads and writes only its latent's DC mode, whatever its atom count (proved; measured on seven rows). At exactly simulable sizes a matched classical core does that job as well as the register; the one candidate regime, long-horizon rollouts under rollout-aware training, did not replicate at four seeds. Next are long-horizon world models for digital twins (Li et al., 2025, Appendix S), where per-step error compounds, and neutral-atom hardware testing whether native Hamiltonian dynamics raise η where classical cores cannot. None of these results is a claim of quantum advantage (Schuld & Killoran, 2022; Cerezo et al., 2025).

AI USE STATEMENT

In this work we used a generative AI coding agent (Anthropic's Claude Code) for code implementation, running and monitoring experiments, computing statistics, producing figures, literature search, and drafting and revising the manuscript text, under the authors' direction and written protocols; and a second agent (Google's Gemini, via the Antigravity CLI) for an independent leakage review of code, a reviewer-style critique of the manuscript, and one validation-only post-processing sweep, whose outputs the first agent checked before use; and, after submission, a third agent (OpenAI's Codex) for an adversarial review of the submitted version and a draft of the protocol for the prospective test (Appendix H), which the authors adopted. The agents also helped propose controls and draft protocol and decision-rule language; the authors directed the hypotheses, benchmark choices, decision rules and kill rules and take responsibility for them. We have reviewed the AI-assisted work: the reported numbers are produced by released scripts from released result files, and the audit rows are cross-checked by an independent script (`paper/judge/judge.py`); citations were checked against the publisher's or arXiv record; core components (simulators, routing, scoring) have unit tests. We take responsibility for the final content of this work, including text, claims and artifacts produced with the aid of generative AI.

ETHICS STATEMENT

The work uses public simulation and measurement datasets under their licenses (RealPDEBench under CC BY-NC, used for non-commercial research) and a benchmark we generated with an open-source simulator; no human subjects or personal data are involved. The application domains (fusion, geothermal and flow modelling) are civilian energy research. All experiments ran on one workstation; negative results are reported in full to discourage selective reporting of quantum benefits.

REPRODUCIBILITY STATEMENT

Section 4 and Appendices A–L specify every protocol, split, metric and statistic. The code release contains the register implementation with unit tests, checkpoint-surgery and fine-tuning code, evaluation scripts for every benchmark, the audit script that regenerates Table 1 from per-seed result files, the Lean 4 sources of Proposition 1, the timestamped pre-registration record, training logs and evaluation arrays, and the package lock file; released checkpoints and datasets are public. The anonymized supplementary material already contains the code, the Lean sources, every per-seed result file behind the allocation, scaling, probabilistic and routing tables, the pre-registration notes, and a script that regenerates those tables.

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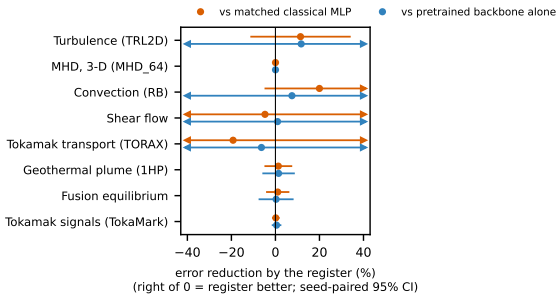


Figure 6: The audit at a glance: every 95% interval of the register's seed-paired gain crosses zero.

Table 4: Matched-control audit, primary rows (three seeds unless noted; lower is better except Sophelio S). Gains are seed-paired relative improvements of the register (positive = better), with the number of seeds on which the register is better and the paired t-test p. Metrics: VRMSE (@8: eighth rollout step), MAE in °C, official score S, NRMSE. CI: paired 95% CI half-width of the register-vs-backbone gain (the effect just significant at $p < 0.05$ given the observed spread; not a power-based MDE). Backbones: FNO (TRL2D arena), released CNextU-net (MHD_64, rayleigh_benard, shear_flow), 1-D spectral (TORAX-Bench), U-Net (heat plume, Sophelio starter), released TokaMind (TokaMark). †Two length-matched seeds (Appendix L). ‡Backbone tuned under the earlier single-seed objective (mixed protocol). On rayleigh_benard and shear_flow, persistence ($\hat{y}_{t+1} = x_t$) scores 0.2024 and 0.0646, below every backbone and adapter in this table; the fix is in Appendix I.

Benchmark	Backbone	+MLP	+Reg.	Reg. vs MLP (%)		Reg. vs backbone (%)	CI (%)
TRL2D arena	0.9567	0.9534	0.8447	+11.4	3/3, .16	+11.7 2/3	53
MHD_64†	0.1897	0.1897	0.1896	+0.0	2/2, .28	+0.0 2/2	0.3
rayleigh_benard	0.4563	0.5275	0.4221	+20.0	3/3, .07	+7.5 1/3	69
shear_flow	0.2686	0.2539	0.2659	−4.7	2/3, .85	+1.0 2/3	99
TORAX-Bench	0.0158	0.0141	0.0168	−19.2	0/3, .40	−6.4 2/3	84
1HP heat plume‡	0.0291	0.0292	0.0288	+1.3	2/3, .44	+1.2 2/3	5.2
Sophelio $S\uparrow$‡	0.7905	0.7848	0.7929	+1.0	2/3, .46	+0.3 1/3	7.5
TokaMark 1-1	0.1509	0.1504	0.1501	+0.2	1/3, .63	+0.5 2/3	1.8
Rows better / worse / tied (sign-test p)				4 / 2 / 2 (p=0.69)		5 / 1 / 2 (p=0.22)	
Random-effects pooled gain [95% CI]				+0.53% [−0.39, +1.45]		+0.04% [−0.00, +0.08]	

A AUDIT DETAILS

The audit script reads every per-seed value from the result files rather than from summaries, fixes one primary row per benchmark in advance, and reports per row the seed-paired gains, paired t, sign counts, d_z, Glass's Δ and the paired 95% CI half-width; across rows it reports a tie-aware sign test, the mean over rows with a t interval, and a DerSimonian–Laird random-effects pool with I^2 and leave-one-out values. Secondary rows: MHD_64 with all three seeds at unequal training length (register vs MLP +4.2%, 3/3, $p = 0.42$; flagged because four of nine runs stopped at about 7.5k of 20k steps and at matched length the arms agree to four decimals with the gate never opening, $|\alpha| \approx 4\times10^{-5}$); TORAX-Bench at fixed untuned hyperparameters (register vs backbone +9.1%, 3/3, $p = 0.14$; vs MLP +1.1%, 1/3); the heat plume at fixed hyperparameters (register vs MLP +0.7%, 1/3); Sophelio at fixed hyperparameters (register vs MLP −0.2%, 1/3); TokaMark with a CNN backbone (one seed; register vs backbone +4.0%); and the TRL2D arena with a shared-link register (+13.3% vs MLP, 3/3, $p = 0.20$). Three errors were caught by the script before reaching a claim: the unequal-length MHD comparison above; a shear-flow published anchor our evaluator does not reproduce (released checkpoint 0.9946 vs published 0.8080); and an inverse-variance pool that rows with identical arms dominate, which is why the mean over rows and leave-one-out values are reported. Register vs MLP leave-one-out pools range from +0.16% to +2.16%, every interval including zero.

B BACKBONE ARCHITECTURES AND INSERTION POINTS

Figure 1 draws the TRL2D instance at layer level. Figure 7 draws the other backbones of the audit (Table 4) and of the comparison with published numbers (Appendix G) in the same way: feature maps as slabs labelled channels and grid size, fully connected layers as bars labelled with their widths, Fourier and transformer layers as blocks. Shapes and parameter counts are traced from the code and the released configurations (`paper/alloc/arch_dump.py`). The register (orange) sits at one latent per backbone. Where the latent is a field it is pooled first (the FNO before its projection, the CNextU-net neck, the 1HP U-Net bottleneck, between the blocks of the TORAX-Bench spectral network); where the latent is a vector it is read whole, with no pooling (the first hidden layer of the Sophelio starter UNetLite, TokaMind's CLS token; §3). Every write-back is zero-initialized and gated, so each hybrid starts at its released or pretrained backbone. The routed co-processor (Fig. 1, bottom) does not depend on the backbone: it reads the backbone output and persistence and writes POD-mode corrections. The vehicle-routing solvers are described in Appendix R.

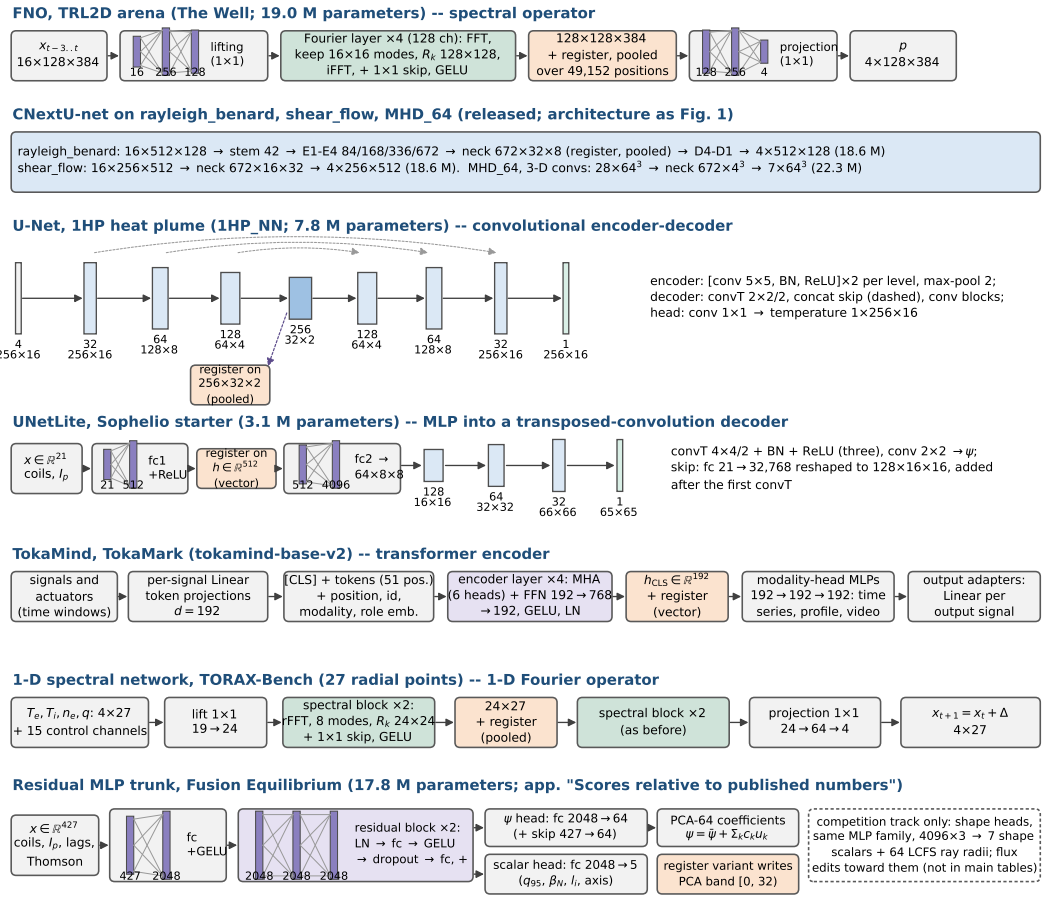


Figure 7: The other backbones at layer level, with tensor sizes, parameter counts and the register's insertion point (orange). The shape heads in the last row's dashed box belong to the competition track only and enter no table of the main text.

C SPECTRAL EVIDENCE AND MACHINE-CHECKED STATEMENTS

Both decompositions conserve energy (Haar relative error 0, FFT shells 2×10^{-7}). With Haar wavelets the backbone misses 11% of the change at the coarsest scale, rising to 18, 26, 32 and 37% at 16, 8, 4 and 2 pixels; the top 30 of 196,416 detail tiles per sample hold 32.9% of the detail

Table 5: TRL2D (180 test windows): where the released checkpoint's error lives, by FFT shell $|k|$, and what the register changes (paired over three Round-1 seeds, relative change of that shell's error energy vs vanilla).

$\|k\|$ shell	error energy share	change energy share	change missed by backbone	register vs vanilla
0 (DC)	0.029	0.022	0.23	$-0.49\% \pm 0.21$ (3/3)
1–4	0.201	0.361	0.08–0.11	within noise
5–16	0.428	0.459	0.11–0.21	within noise
17–32	0.231	0.116	0.34	within noise
$\geq$33	0.111	0.042	0.44–0.45	within noise

error energy, eleven times the DC ceiling, and a zero-parameter router that ranks tiles by the backbone's own predicted change captures 9.4% at K=30 (about 1,650 times a random choice); learned routers beat it only at $K \geq 3{,}000$. A routed register on TRL2D is a pilot (Section 6).

Machine-checked statements. Lean 4 with Mathlib, standard axioms only (`propext`, `Classical.choice`, `Quot.sound`), no `sorry`: `pool_blind` and `pool_blind_mode` (mean pooling is invariant to zero-mean perturbations, in particular non-DC Fourier modes); `broadcast_orthogonal`; `band_energy_split` and `residual_after_band_removal` (Parseval split and the exact residual after a perfect correction on a coefficient set, the ceiling of any K-tile corrector); `pointwise_min_optimal` and `swap_improves` (top-K by energy is the optimal budgeted assignment); `reupload_degree` (an L-layer re-uploading circuit's spectrum lies in $[-L, L]$); `vdw_sensitivity_anti` (geometry acts as frequency modulation); `lattice_sep`; and `dip_inplane`, `dip_axial`, `dip_magic` (signed dipolar couplings require an out-of-plane coordinate); and, for the allocation study (§5.5), `gain_le_ceiling` (in any orthonormal basis, a correction supported on a coefficient set T, with arbitrary coefficients, lowers the error energy by at most the error energy on T), `ceiling_mono` (nested allocations have nested ceilings), and `pooled_gain_le_dc` (a spatially constant correction c lowers $\sum_i e_i^2$ by at most $(\sum_i e_i)^2/n$, whatever core and however many atoms produced c); and, for the frequency-localized router and the persistence gate (Appendix G), `band_span_le_energy` (Bessel inside a band: the energy on orthonormal band-localized modes never exceeds the band's energy, so captured$_b$ $\leq$ span$_b$ $\leq$ full$_b$ per band and window), `gate_energy_exact` and `gate_coeff_exact` (the gated residual's energy and every mode coefficient are exact functions of three band statistics and two coefficients, for any gate value), and `gate_opt_le` with its specializations `gate_opt_le_persistence` and `gate_opt_le_trunk` (the best scalar gate per band is no worse than persistence or the backbone). The standard step that turns a parameterized circuit into a truncated Fourier series is not formalized. A separate file checks algebraic consequences of a proposed configuration-aware error model ($E = E_\infty + AP^{-\alpha}(1 - g\,u) + \ldots$ with utilization $u \in [0, 1)$): improvement with utilization holds only when the signed gain $g \geq 0$, which is an empirical question the audit answers with a null.

D MOVING THE INSERTION: PROTOCOLS

RealPDEBench. Airfoil (NACA4418) real PIV velocity, 20 input and 20 output steps at 64×128; the benchmark's public splits (5,133 validation and 5,140 test windows over 98 trajectories), normalizer and metric code, with the full-set metric computed in exact chunks. The corrector reads the lowest $3 \times 4 \times 4$ space–time modes of the backbone's prediction and of the last input window, standardizes them on the training split, passes them through a zero-initialized gated core (none, linear, matched MLP, or 3×10-atom register; the MLP and register share input and output maps), and writes a correction to the same modes; it trains on training windows, early-stops on validation relative L_2, and scores test once per arm and seed. The corrector does not exceed the released paper's reported numbers, which are not reproducible on the current data release; the comparison is to the released checkpoint evaluated with the benchmark's code.

Fusion Equilibrium Challenge. Full-data pipeline (7,041 DIII-D shots, 1.56M frames; magnetics and Thomson profiles; an extraction unit error placed the intended -200 to $+20$ ms magnetics lags

within 0.2 ms, so these models see no temporal context, Appendix G), a PCA-64 flux decoder whose ground-truth projection scores $S = 0.9991$, and a stage-1 architecture search that fixed the frozen backbone. The stage-2 search over couplings (latent or spectral-routed, register size and graph, $\leq$30 atoms) ran on VALID only; its top three configurations were re-run with three seeds beside matched MLP twins. Register vs MLP by configuration: 10-atom ring, learning rate 2.4×10^{-3}: 0.9878 vs 0.9876 ($p = 0.43$); same register at 1.6×10^{-3}: 0.9878 vs 0.9868 ($p = 0.030$, Bonferroni 0.09); 5-atom ring: 0.9879 vs 0.9876 ($p = 0.41$). After the single registered test scoring (0.9910 on our held-out split of training shots; the development leaderboard, which scores the public test fold, stood at 0.9932 in August and 0.9959 on September 25), a post-processing sweep on VALID only (shrinkage of high-order PCA coefficients, per-shot temporal smoothing, outlier filters) raised the full-VALID score of the entry ensemble from 0.99127 to 0.99143. That is 1.7 times the estimated ensemble seed SD, below the threshold we set in advance (3 times) for spending a second test scoring, so no second test was run. The same selection rule, registered before the 3-D register results, was then applied to two further pools and each selection scored once on the held-out split: classical trunks with the 3-D register models (S 0.99100, six members) and with their 3-D MLP twins (0.99099, seven). They tie each other and the classical-plus-MLP entry (0.99102); the 3-D register selection is 0.0005 above the earlier 2-D register selection (0.99046), and the classical-only selection scores 0.99097.

E ALLOCATION STUDY: PROTOCOL, PER-STRATEGY RESULTS AND THE ANALOG–DIGITAL PROBE

Register (fixed backbone)	Atoms	VRMSE@8
FNO backbone alone	0	0.9567 ± 0.2554
+ Rydberg register	5	0.8427 ± 0.0709
+ Rydberg register	10	0.9332 ± 0.2166
+ Rydberg registers	20	0.9269 ± 0.2792
+ Rydberg registers (3×10)	30	0.8447 ± 0.0511
+ matched MLP (control)	—	0.9534 ± 0.1354
+ 30 atoms, conditioned geometry	30	0.8951 ± 0.1367
+ 30 atoms, mean-field pilot	30	0.8393 ± 0.0463

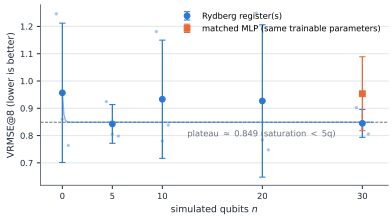


Figure 8: Rollout accuracy versus register size at fixed backbone, data and step budget (sprint arena, 3 seeds; amber: matched MLP). The 0- and 30-atom arms ran first; the interior arms and the mean-field pilot were pre-registered before execution. Points above 10 atoms add registers of 10. A constant fits the $n \geq 5$ points (a saturating fit puts any rise below five atoms).

Qubit count and inertness. In a fixed-budget arena (FNO backbone, 2,000 steps), accuracy is flat from five qubits: 5 atoms reach 0.8427 ± 0.0709 against 0.8447 ± 0.0511 for 30 (Figure 8). The trained register is inference-inert (product-state surrogates reproduce it within 2.4×10^{-5}), so any gain arises in training (mechanism tests: Appendix J).

Caches and exactness. Each frozen backbone predicts every window once (fp32; the convolutional and spectral backbones have no layers NVFP4 applies to). Training windows are a uniform random sample of 3,000 (700 on the heat plume); validation and test windows are uniform samples of all windows or the full split. For every window we store the per-channel residual energy, the metric denominators, and, for 320 tiles per rule (proxy, static, random, oracle), the Haar coefficients of $p - x_t$, p, x_t, $x_t - x_{t-1}$ and r. Because the transform is orthonormal, the corrected per-channel error is $E_c - \sum_{i \in S} \left(r_{ic}^2 - (r_{ic} - c_{ic})^2\right)$ exactly, so VRMSE and relative L_2 are computed without reconstructing fields; captured $\leq U$ held on every cell. On the heat plume, the "state" is the training-mean temperature field and the "change" is the permeability input, since the task is not autoregressive; its study metric is VRMSE of the normalized temperature, not the benchmark's MAE.

Strategies and corrector. A *tile* is one Haar scale, orientation and position, all channels together. *Pooled*: global statistics in, one broadcast constant per channel out. *Static*: the K tiles with the largest mean training error. *Proxy*: top-K by the energy of the backbone's own predicted change. *Learned*: a scorer trained to rank residual energy. *Gain*: a scorer trained on the co-processor's own realized per-tile gain. *Random*. Atoms sit at their tiles (x, y from position, z from scale); detunings

come from tile features and global context. Each tile receives $c_i = s_i(W_{\kappa(i)} f_i + A\, o_i)$ with a per-(scale, orientation) linear map W shared by every arm, the core's readout $o_i = (\langle Z_i \rangle, \langle X_i \rangle, \langle Y_i \rangle)$ and a zero-initialized A. With $A = 0$ (the registered linear arm) the corrector alone gains 28.8% on rayleigh_benard (proxy) and 26.3% on shear_flow (learned), test, three seeds; the cores' further 7.5 points on rayleigh_benard come from its heaviest-error windows (ranked by the backbone's error, the top 10% carry 96% of them; the median window changes by 10^{-4}), and on the other rows, at the listed tile strategies, the register adds at most 0.8 points (heat plume; MLP 1.0; 1.6 at the heat plume's learned strategy). Input-adaptive routing adds 3.9 points to the linear corrector on shear_flow but none on rayleigh_benard, where only the cores gain from it (3.4 register, 3.5 MLP). A registered replication with five new seeds (test windows reused) finds no register edge over the matched MLP on the convection-onset windows ($t_0 < 10$; MLP minus register -0.09 to $+0.06$ VRMSE, positive on 3/5 seeds), while both beat the classical mean-field twin there by 0.55–0.85 on every seed. The amendments: global context for tile co-processors after the first cells; planetswe excluded because its released checkpoints do not reproduce.

Co-processor. Tile features are scale-free (coefficients divided by the tile's own scale s_i, clipped), plus log scale, scale and orientation one-hots, normalized position and the window's global statistics. Registers are groups of 10 atoms in Morton (Z-order) order of their tiles; each atom's detuning per layer is $b_{\ell i} + \pi \tanh(w_\ell^\top f_i)$ (weights shared across atoms), positions are tile coordinates divided by learned per-axis blockade scales plus a learned scale-plane separation, and four layers of the propagator of §3 are applied. The DeepSets MLP sees $[f_i, \bar{f}]$ and has the nearest width to the register's parameter count (e.g. 316 vs 271 on TRL2D, 156 vs 165 on the heat plume). Scorers are two-layer MLPs trained with a listwise loss over each window's proxy shortlist, on residual energy (learned) or on the proxy-trained co-processor's realized per-tile gain (gain). AdamW, learning rate $3{\times}10^{-3}$, batch 64 windows, up to 40 epochs with patience 6 on validation; the loss is the negative captured share. A learning-rate check on TRL2D (10^{-3} to 10^{-4}) left the TRL2D null unchanged.

Analog–digital probe. Groups of 10 atoms with features $f_i \sim U[-1, 1]^4$ on random sites of a $4{\times}4{\times}2$ grid; targets per atom $\sin(\omega\, f_i^\top u)\cos(\omega\, \overline{f^\top v})$ (analog), $\mathrm{sign}(f_i^\top u)\,\mathrm{sign}(\overline{f^\top v})$ (digital); 20k/5k/10k groups, Adam $3{\times}10^{-3}$, 3,000 steps, best-validation checkpoint, test R^2, three seeds. Registered rule: an analog edge requires the register to beat its matched MLP on 3/3 seeds at ≥ 3 of the five ω. Test R^2, mean of three seeds (*exploratory arms added after the $\omega \leq 4$ results; MLP: nearest width to the register's 84 parameters, 75; MLP$^+$: next width up, 93):

core	ω=0.5	ω=1	ω=2	ω=4	ω=8	ω=16	digital
register	0.999	0.998	0.990	0.873	0.000	-0.000	0.067
mean-field	0.803	0.971	0.990	0.642	-0.000	—	0.226
MLP	1.000	0.999	0.979	0.953	-0.000	-0.000	0.719
MLP$^+$	1.000	1.000	0.989	0.956	-0.000	-0.000	0.920
MLP×4	1.000	1.000	0.986	0.989	0.012	-0.000	0.954
register, linear enc.*	—	—	—	0.874	-0.000	-0.000	—
sine-MLP*	—	—	—	0.988	-0.000	-0.000	—

Outside physics, on 22 TDC ADMET endpoints the register trails its matched MLP on 13 of 22 (it leads on 9).

Caco-2 leader audit. The Caco-2 leader (CaliciBoost, MAE 0.256) reproduces exactly with its released code (0.2560 ± 0.0060, five seeds), but only with its hard-coded per-seed feature subsets and its GPU tree builder (CPU histograms: 0.271). TDC's training and test sets do not change with the seed, yet the subsets do, and the code has no validation split. Without selection the same model scores 0.294 ± 0.004; choosing the same number of features by permutation importance inside the training data, as a method must, gives 0.308 ± 0.010. We list 0.256 as published, but it cannot be re-derived from training data alone; our best, 0.276 (MapLight features and CheMeleon embeddings, CatBoost), is ahead of both re-derived values.

Positive control: quantum-generated targets. Registered before any run, in the same harness: each atom's target is $\langle Z_i \rangle$ after a 30-step Trotterized interacting Rydberg quench (constant drive Ω=1.5, input-set detunings $\delta_i = 2 f_i^\top u$, $V_{ij} = r_{ij}^{-6}$ with nearest neighbours at $V \approx 1.8$). The teacher's

depth and fixed geometry lie outside the trained 4-layer ansatz. After a design critique, and before any register result, we added message-passing cores that see the teacher's exact pair interactions. Test R^2 (three seeds): mean-field register 0.671, register 0.433, GNN 0.333 and 0.357 (4×), MLPs 0.19–0.20. The register beats every MLP and GNN on every seed, so the harness separates cores sharply when the target calls for it. It loses to its mean-field twin on every seed, however, so the registered rule is not met. What helps on these targets is the Rydberg structure (geometry and r^{-6} interactions), which the classical mean-field model carries. At ten atoms and this training budget, entanglement adds nothing.

F COMPUTE COST

All runs used one NVIDIA DGX Spark (GB10 Grace Blackwell chip, 128 GB of unified CPU–GPU memory, Arm64 host); no cluster or cloud compute was used. Median wall-clock per training step on the TRL2D neck insertion (released CNextU-net, $4 \times 16 \times 128 \times 384$ input, forward, backward and AdamW in float32, 25 steps after 5 warm-up, two repetitions, GPU shared with other jobs): backbone 504 ms, matched MLP 511 ms, 30-atom register 595 ms. The exact statevector propagator is the only added cost; inference with a trained register can use its product-state surrogate at the MLP's cost because the trained register is inference-inert.

Qubits are not parameters. Every register–MLP comparison sizes the MLP from the register's parameter count, never at one parameter per qubit (exactly in the neck audit; in the routed co-processor the integer hidden width leaves the MLP smaller, as below). A 10-atom group evolves a $2^{10} = 1{,}024$-amplitude state, but only its detuning maps, drives, evolution times and coordinates are trained, and it returns three expectation values per atom, not its amplitudes. In the routed co-processor on TRL2D at 30 qubits the register core has 244 parameters (241 trainable; its three length-scale parameters are frozen) and its MLP 199, which leaves the MLP two hidden units per atom. The qubit axis of Appendix P is therefore a different axis from the parameter counts of classical world models: 30 to 50 qubits add a few hundred trainable parameters to a backbone of 18.6 million (the TRL2D CNextU-net). That the register still ties an MLP with no more trainable parameters says these tasks, at these sizes, do not draw on its larger state space.

G SCORES RELATIVE TO PUBLISHED NUMBERS (ALL CLASSICAL LIFTS)

Table 6: Our best numbers against published ones. Every gain is attributable to classical components (the register effect on the TRL2D long-horizon row did not replicate at four seeds, §5.2) (fine-tuning, correctors, ensembles, edge-weighted training); the register's contribution is measured in Tables 1–2. Ensembles are labelled as such.

Benchmark	Published best	Released checkpoint, our evaluator	Persistence	Ours (component)
TRL2D one-step VRMSE ↓	Flowers 0.1930 (CNextU-net 0.1956)	0.1934	0.5761	**0.1909** (gated fine-tuning, 3-backbone ensemble; plain 3-seed ensemble 0.1929; Flowers' conditioned 1→1 protocol: 0.1907)
TRL2D rollout 13:30 ↓	CNextU-net 1.01	1.0132	—	0.9528 (TD fine-tune with register, 3 seeds; 0.9660 with the registered fourth)
RealPDEBench foil rel. L_2 ↓	released U-Net	0.00948	—	0.00931 (linear corrector + 0.2 U-Net-real, VALID-weighted)
TokaMark 1-1 ↓	TokaMind-Scratch 0.140 (FT-Base 0.150)	—	n/a	0.1334 (3 Scratch-recipe runs, equal weight, VALID-selected after the registered 4-member ensemble, 0.1400; one run 0.1379)
shear_flow ↓	PDE-FM 0.0345 (PhysiX 0.0700)	0.9946	0.0646	**0.0207** (gated classical part + 30 routed qubits, 3 seeds; classical part alone 0.0236; with a metric-aligned loss 0.0197, 2 seeds, matched MLP 0.0199, seeds split 1–1)
rayleigh_benard ↓	PDE-FM 0.0415 (PhysiX 0.1470)	0.6679	0.2024	0.0403 (gated, metric-aligned fine-tuning; equal-weight 3-backbone ensemble; best single model 0.0425; backbone alone 0.4563; 8-pass x-symmetry average 0.0366[†])
Fusion Equilibrium Ch. 1 S ↑	0.9959*	—	n/a	TEST 0.9949 (VALID-selected 7-member ensemble 0.9910, VALID 0.9913; with registered shape-head post-processing and a ψ-specialist ensemble as described below; still below the best)

*Fusion Equilibrium: best development-leaderboard entry on the public DIII-D test fold, September 25, 2026 (0.9932 in August). Our TEST is a held-out split of the training shots, not the same test set. MHD_64's released checkpoint evaluates to 0.5447 per-channel VRMSE, not the published 0.1633 (our gated, fine-tuned backbone: 0.1725 on the full test split; current best 0.1160, EddyFormer. A Flowers model trained from scratch with the authors' schedule scores 0.1216 (their 0.1165); with the divergence-free and conserved-mean projections 0.1191, after a 5-epoch fine-tune on the VRMSE itself 0.1172, and blended with the gated CNextU-net by channel and band (weights fit on TRAIN) 0.1154, which we call parity; symmetry averaging, 8× inference, gives 0.1035[†]. In the CNextU-net model, its error is regime-level, not a tail: over the two fine-tuned CNextU-net seeds trained to full length the five super-Alfvénic test runs, M_A=2, score 0.217–0.218 against 0.136–0.137 at M_A=0.7 (61% of the error), flat in time, and the worst 10% of windows hold 14% of the mean, which leaves routed modes little to reach), and shear_flow's to 0.9946, not 0.8080; on shear_flow we therefore compare with the foundation models PDE-FM (Soares et al., 2025) and PhysiX (Nguyen et al., 2025), which report full official test splits without ensembles or test-time augmentation, and not with the released checkpoint. Our classical part alone (0.0236, full test split) is below PDE-FM's 0.0345; on rayleigh_benard our ensemble (0.0403) is within 3% of PDE-FM's single model

(0.0415), which we call parity, not a win; our best single model scores 0.0425 (its first test score), 2.3% above it. Averaging each model over the exact symmetries of the periodic cell (reflection $x \to -x$ with $u_x \to -u_x$, times four periodic shifts; $^{\dagger}8\times$ inference, registered before its test score) gives 0.0383 for the single model and 0.0366 for the ensemble (-9.8% and -9.1%, paired over test trajectories, 95% CIs within ± 0.8 points); PDE-FM reports a single pass. The gap between the register and the matched MLP neck (one checkpoint each, seed 1337: 0.0425 vs 0.0438, -3.1%) shrinks to -0.8% under the same averaging (0.0383 vs 0.0386); it is one checkpoint pair, not a training replication (the multi-seed tests are in Appendix P). Training protocols differ (per-dataset fine-tuning here, cross-physics pre-training there) and are not normalized. Persistence ($\hat{y}_{t+1} = x_t$, full official test splits, the benchmark's own mean VRMSE) is a necessary baseline on the Well rows: it scores 0.2024 on rayleigh_benard and 0.0646 on shear_flow, below The Well's published baselines, PhysiX on shear_flow (0.0700) and every backbone we trained (0.36–0.57 and 0.22–0.34), though not PDE-FM, although the backbones win 84% and 54–63% of the individual windows; their mean is dominated by a heavy tail of low-variance windows. On TRL2D (0.5761) and MHD_64 (0.4183, beaten on every window) the backbones are far better than persistence. We therefore make no state-of-the-art statement from backbone numbers on rayleigh_benard or shear_flow. With the band gate and metric-aligned fine-tuning (Appendix I), rayleigh_benard reaches 0.0403 on the full test split, a fifth of persistence. This reflects metric alignment, not better convective dynamics: the convection-onset windows ($t_0 < 10$, 3.6% of test windows) carry 39% of the backbone's mean VRMSE (backbone 4.86, persistence 0.038 on them), and the classical gate leans on persistence per band and window where the backbone is unreliable. Recent foundation and compact models on The Well (McCabe et al., 2025; Sankaran et al., 2026) report the *median* one-step VRMSE over windows, not the benchmark's mean. Because the mean is dominated by the heavy tail discussed above, their numbers are not comparable to the mean-based scores in this table.

Where TokaMark 1-1 is lost. Training TokaMind from scratch with its published recipe gives 0.1379 in our evaluator (one seed), matching the reported 0.140; the registered equal-weight ensemble of this model with three fine-tuned FT-Base seeds gives 0.1400: the FT-Base members (0.150–0.151 each) hold three of the four equal weights and dilute the Scratch model. We report this registered ensemble as is. After it, a stratified VALID diagnosis (534 shots) selected the equal-weight ensemble of the three Scratch-recipe runs below (vanilla, MLP neck, register neck; one seed), registered before its TEST score: 0.1334, -3.2% against the single run (paired over 1,020 test shots, 95% CI 2.9–3.6%, 829 shots better); adding the three FT-Base runs with per-feature weights fit on TRAIN gives 0.1332, not resolved from it (-0.17%, CI -0.41 to $+0.07\%$). It is post hoc with respect to the registered ensemble and credits neither neck. With the neck swapped at the same slot on the Scratch recipe (one seed each), the register neck gives 0.1404 and the MLP neck 0.1391: neither helps this backbone. On the FT-Base recipe the 3-D tethered register (full recipe, one seed) scores 0.1512 against 0.1500 for its matched MLP (2,178,573 and 2,178,411 parameters) and 0.1504 without a neck; the MLP is lower on 13 of the 15 signals (one seed, descriptive). The registered ensembles tie: 0.1400 (vanilla), 0.1400 (register neck), 0.1402 (MLP neck). Per target, our 3-seed register-neck FT-Base ensemble (0.1454 overall) beats TokaMind-Scratch (0.140; per-target values from its Table A.2 (Boschi et al., 2026)) on the plasma geometry: x-point z 0.0201 vs 0.022, x-point r 0.1654 vs 0.170, magnetic-axis z 0.0725 vs 0.075. The whole gap is in the three pressure ratios: β_{pol} 0.1655 vs 0.153, β_N 0.1687 vs 0.157, β_{tor} 0.1680 vs 0.157. The vanilla and MLP-neck ensembles show the same pattern. By shot length, the mean per-shot NRMSE is 0.172 on the shortest quarter of test shots ($\leq$218 windows, mostly early terminations) against 0.131–0.141 on the rest, so the gap is concentrated in short, disrupted discharges. It is not a learnable bias. On the test windows, the share of the β squared error explained by its mean within bins of shot length $\times$ time in shot is at most 0.6% (an in-sample upper bound for any additive correction on those bins), and the reconstructed β labels jump 1.5–9.6$\times$ more than the predictions between consecutive windows. We read this as noise in the EFIT labels near termination and fit no correction; the rule was fixed before the numbers.

Where the Fusion Equilibrium score is lost. On full VALID, $1 - S = 0.0088$ for the entry; the consistency term (shape scalars derived from the predicted flux: elongation, internal inductance, volume, axis position) carries 75% of it, the LCFS distance 17%, ψ 2%. Two input errors surfaced in a stratified diagnosis: our magnetics lags were sub-millisecond (above), and on the 20 kHz shots (70%) the challenge's shared plasma-current time vector reads $|I_p| < 5\%$ of its peak at 57–100% of EFIT frames, which cannot hold during a plasma; anchoring it to the magnetics clock removes this

(pre-breakdown $|I_p|$ median 0.2% of peak). A matched retrain of the entry's best trunk (two seeds, 180 stratified VALID shots, official S) registered before running shows that neither correction helps: the re-anchored current alone lowers S by 0.0011–0.0017 (95% CIs below zero); with the true-ms lags as well, pooled ψ and scalar-head R^2 improve on both seeds but S changes by -0.0006 and -0.0005 (CIs span zero), because the flux-derived shape consistency worsens; a second registered pass (an explicit acquisition-rate input; lags floored at -50 ms) lowers S further. The entry is therefore not retrained, and the gap to the best development-leaderboard entry (0.9959) stays open.

Where the Fusion Equilibrium score is recovered. Most of the consistency loss comes from extraction, not the flux map: the true ψ projected on the 64 PCA components scores $S = 0.9991$, and our predicted map has $R^2_\psi = 0.9997$, yet the shape scalars derived from it lose 3%. The scorer reads the magnetic axis as a sub-pixel critical point (the Z_{axis} spread, 0.043 m, is below one 0.05 m cell) and the boundary through the X-point, so small flux errors near critical points are amplified. Direct heads trained on the scorer's own extraction of the training-split flux (from its stored 256-component form; no validation or test labels) beat the shape extracted from our predicted flux on 180 stratified VALID shots: R^2 0.991 vs 0.978 for Z_{axis} (the trunk's own head), 0.965 vs 0.933 for elongation, 0.987 vs 0.966 for volume, boundary radius error 7.1 vs 9.9 mm. Registered inference-only edits move the predicted flux so that the scorer's extraction lands on the heads. The plasma core is shifted (Gaussian window, zero at the boundary) until the scorer's O-point finder returns the head axis. The flux is warped radially about the axis so that the extracted boundary meets the head radii, leaving the far field fixed. An area-preserving elongation about the boundary's centre matches the scorer's elongation to its head. Finally the interior flux profile is remapped toward the internal-inductance head, keeping the axis, boundary level and boundary gradient. Each step was picked on VALID and scored once on our held-out TEST split (373 shots): S 0.99102 $\rightarrow$ 0.99155 (axis), $\rightarrow$ 0.99159 (inductance; a weak replication, $+0.00003$ against $+0.00018$ on VALID), $\rightarrow$ 0.99251 (boundary), $\rightarrow$ 0.99291 (heads retrained on all training frames with gradient clipping, after unclipped long runs diverged in 2 of 3 seeds), $\rightarrow$ 0.99323 (elongation) and $\rightarrow$ 0.99339 (heads trained twice as long with the seven scalars weighted equal to the 64 boundary radii) $\rightarrow$ 0.99343 (six head seeds) and $\rightarrow$ 0.99369 (flux from three ψ-only trunks, trained without the scalar loss, each step re-scored on a GPU port of the post-processing that matches the CPU one under the official scorer) $\rightarrow$ 0.99490 (inputs with the plasma current re-timed on 20 kHz shots and lags in true milliseconds, for the ψ trunks, the q_{95}/β_N trunks and the heads, plus a lateral warp of the top and bottom points toward the triangularity heads; the warp alone adds $+0.00005$). Elongation R^2 rises from 0.931 to 0.959, volume from 0.968 to 0.984, and the boundary distance falls from 0.0146 to 0.0119. Dedicated q_{95}/β_N heads, finer boundary sampling at fixed head quality, and a triangularity shear did not help. The leaderboard file stores ψ in float16, as the starter kit recommends; on VALID that alone costs 0.0004 (the same with and without the edits), so float32 storage is the exact fix.

H CONFIRMATORY AND EXPLORATORY CLAIMS

Every final configuration was scored on TEST once, but most protocols were designed after earlier TEST results on the same rows were known (new seeds or budgets on the same test windows included), so TEST was reused adaptively across amendments and only the first audit round is confirmatory. Table 7 labels each claim. *Confirmatory*: the decision rule and the TEST sample were fixed before any TEST result on that row. *Registered, TEST-informed*: the rule was fixed before the run, but the design followed earlier TEST results on the same rows; these claims are exploratory. *Descriptive*: no decision rule. A prospective test on benchmark rows not used elsewhere in this paper (active_matter and gray_scott_reaction_diffusion; TEST downloaded only after the prediction, the 24 trained models and the rule were locked and hashed; each model scored once) predicted from VALID that error-proxy routing beats pooled insertion in both the register and a matched MLP core. It held on both rows: mean seed-paired TEST gains 12.3% (register) and 14.1% (MLP) on active_matter, 15.2% and 24.8% on gray_scott (trajectory-bootstrap 95% intervals above zero). The MLP gained more than the register on both rows, so this supports this operational proxy-insertion strategy on these rows, not a quantum advantage; it does not isolate routing from the insertion architecture, and the gain is concentrated (two of 21 active_matter configurations give about 80% of the absolute improvement; gray_scott's gliders about 90%). Post hoc, on the gray_scott V2 half of VALID an earlier, separately trained linear-only tile corrector of the same recipe scored below the register-based corrector on all three seeds, and removing the core term from the frozen register

checkpoints also lowered V2 error on all three seeds. On gray_scott the proxy arm has about 60% fewer co-processor parameters than pooled; on active_matter it has about 53% more.

Table 7: Status of each claim. Rows in the order of the main text.

Claim	Status	Where
Pooled neck register ties its matched core on eight benchmarks	confirmatory	§5.1, Table 1
TRL2D long-horizon edge does not replicate at a fourth seed	TEST-informed (negative)	§5.2
Frequency gate beats persistence and the backbone	TEST-informed	§5.4, Appendix I
Routing gains 26%/36% vs 7% pooled (linear corrector alone 26%/29%); matched cores gain as much	TEST-informed	§5.5, Table 3
Error falls with the budget; bounds at 50, 100, 200 qubits hold	TEST-informed (U from test labels)	§5.5, Appendix P
Register beats a smaller graph-blind MLP (199 vs 244 parameters) at 100 qubits on shear flow (5/5 new seeds), not established against a graph-aware GNN (239) or a wider MLP (297)	registered replication, TEST-informed	Appendix P
Closed form $G(K)$; captured share $\leq U$ (an algebraic check)	descriptive	§5.5
Tested per-input geometry, tethers and 3-D layout did not meet the registered gain rule	TEST-informed (negative)	Appendix P
3-D register vs its matched MLP per benchmark	TEST-informed; no two-sided test passes	Appendix P
Probabilistic hybrid: lower proper energy score than the deterministic hybrid and a VALID-tuned dressing (shear_flow, rayleigh_benard; 3/3 seeds; matched MLP equal)	registered	Appendix Q
Scores against published numbers	descriptive	Appendix G
Routing gains are predictable from validation data on unseen rows	confirmatory, pending	registered

I Evaluation-error analysis: persistence, a frequency-aware gate, and metric-aligned fine-tuning

Where the Well backbones fail. The benchmark's score is the mean over windows of a per-window, per-field VRMSE, which divides each window's error by that window's variance. On rayleigh_benard and shear_flow this mean is dominated by a heavy tail of low-variance windows (early convection onset; slowly evolving shear layers). There the backbones, trained with plain MSE on normalized fields, give those windows almost no weight. Persistence ($\hat{y}_{t+1} = x_t$) is therefore a strong baseline: on the full official test splits it scores 0.2024 and 0.0646, below The Well's published baselines (0.6566, 0.8080) and every backbone we fine-tuned, though not PDE-FM (0.0415, 0.0345), although the backbones win 84% and 54–63% of individual windows. A per-window oracle choice between backbone and persistence would score 0.044–0.048 and 0.036–0.039. On TRL2D (0.5761) and MHD_64 (0.4183) persistence is far behind.

A classical, frequency-aware gate. A related classical design in Hybrid Neural World Models (Lakshmanan & Chopra, 2026) gates a neural surrogate against a reference-solver fallback using an error map that detects sharp dynamical events (shocks, fronts). In our architecture, the fallback is persistence, evaluated per window and Fourier octave; rather than abandoning surrogate throughput via an expensive numerical solver fallback, QUOTA routes the remaining frequency-localized residual modes directly to an agile quantum (or matched classical) co-processor. We split the backbone's increment over persistence into five dyadic ℓ_∞ Fourier shells P_b (orthogonal projectors summing to the identity) and predict $\hat{y} = x_t + \sum_b \alpha_{c,b} P_b\big(p(x) - x_t\big)$ per field c. With α fitted in closed form on training windows (the *blend*) or predicted per window from causal features (the *gate*: band energies of $p - x_t$ and of the last input increment, normalized by the last input frame's variance), the gated residual's energy and every mode coefficient are exact functions of cached band statistics (`gate_energy_exact`, `gate_coeff_exact`; the per-band oracle gate is no worse than per-

sistence or the backbone, `gate_opt_le`). The blend keeps the backbone's low bands and falls back to persistence for the high bands of the velocity fields ($\alpha \approx 1$ at low k, 0–0.1 at the top octave). A first version normalized the gate's features by the *target* frame's variance. We caught this leak, archived those runs, and reran with causal features. Every number below is causal; the change moved results by less than 1%.

The routed register after the gate. The residual that remains is written in band-wise POD modes (POD computed inside each band, so each qubit carries a signal of known band, and per band captured$_b \le$ span$_b \le$ full$_b$ holds for every window, `band_span_le_energy`). A learned router sends 30 modes per window to the register, and a classical head takes the rest. Without supervision, the router left the register nearly idle: it put 92% of its slots in the top octave, which holds almost none of the reachable error. Supervising the router toward each window's largest reachable error moved 80% of the slots to the lowest band. Table 8 reports the matched cores (register, GNN on its own learned graph, MLP) with the supervised router. With supervision, routing matches the allocation thesis: the qubit-routed modes carry most of the co-processor's capture, in the band where the reachable error is (Table 9). But the matched classical cores, each behind its own separately trained router of the same design, capture as much: the cores tie (the register's and MLP's routed sets overlap by 40% on shear_flow, seed 1). We attribute the gains to the gate, the routing and the classical head.

Table 8: Evaluation-error analysis on the Well rows (VRMSE, lower is better). Published: best published number (PDE-FM on shear_flow and rayleigh_benard, Flowers 4→1 on TRL2D, Eddy-Former on MHD_64). Backbone: the fine-tuned (or released) CNextU-net. Blend: TRAIN-fitted band blend with persistence. Gate: per-window causal band gate. The last three columns stack the routed co-processor (30 atoms or matched classical cores) on the gate; mean ± SD over seeds. Full official test splits except † (stratified test sample, every trajectory × 8 time bins, 4 windows each; validated against full-split per-window scores to within 2%). The Well's CNextU-net claims 0.1633 on MHD_64; its released checkpoint evaluates to 0.5447.

Row (backbone)	Published	Persistence	Backbone	Blend	Gate	+Register	+GNN	+MLP
shear_flow (register neck)	0.0345	0.0646	0.2297	0.0451	0.0272	0.0201 ± 0.0002	0.0205 ± 0.0002	0.0200 ± 0.0003
rayleigh_benard† (register neck)	0.0415	0.2024	0.4321	0.1420	0.0420	0.0413	0.0415 ± 0.0003	0.0416 ± 0.0004
rayleigh_benard† (vanilla)	0.0415	0.2024	0.4252	0.1438	0.0436	—	—	—
TRL2D (register neck)	0.1930	0.5761	0.1931	0.1924	0.1921	0.1922	0.1922 ± 0.0001	0.1922
TRL2D (released)	0.1930	0.5761	0.1934	0.1926	0.1924	0.1923 ± 0.0001	0.1923 ± 0.0001	0.1923 ± 0.0001
MHD_64 (vanilla)	0.1160	0.4183	0.1771	0.1761	0.1735	0.1733	0.1733	0.1733

Table 9: Where the register's work goes, per Fourier band, after the gate (route-supervised router, seed 1). Shares of each window's error before the gate, averaged over test windows. full$_b$: error in band b after the gate. span$_b$: part of it on the band's POD modes. Then what the qubit-routed modes and the classical head capture, and the share of the 30 qubit slots that land in the band. The chain captured$_b \le$ span$_b \le$ full$_b$ is checked on every window; the largest violations are fp16 rounding.

Row	Band	full$_b$	span$_b$	Captured (qubits)	Captured (classical)	Qubit slots
shear_flow	b_0	0.3891	0.3433	0.2167	0.0419	95%
	b_1	0.0444	0.0186	0.0050	0.0008	2%
	b_2	0.0360	0.0080	0.0005	0.0000	3%
	b_3	0.0297	0.0016	-0.0000	-0.0004	0%
	b_4	0.0342	0.0006	0.0000	-0.0002	0%
	max violation: captured≤span -1.5e-10, span≤full -5.3e-08					
rayleigh_benard	b_0	0.0476	0.0359	0.0186	0.0011	81%
	b_1	0.0180	0.0047	0.0008	0.0001	12%
	b_2	0.0151	0.0017	0.0009	0.0000	2%
	b_3	0.0191	0.0008	0.0007	-0.0000	1%
	b_4	0.0267	0.0008	0.0004	0.0000	3%
	max violation: captured≤span -3.2e-09, span≤full -6.0e-07					

Metric-aligned end-to-end fine-tuning. The fix belongs in training, not only in stacking. We fine-tune the whole stack end to end: backbone (vanilla, MLP neck, or register neck), gate, and the benchmark metric itself as the loss. Selection uses a stratified validation sample; the selected configuration is scored once on the full test split (Table 10).

Table 10: End-to-end fine-tuning with the band gate (seed 1). Loss: VRMSE (the benchmark metric) or MSE. Schedule: joint from the start, or exact gate pre-training on 3,000 frozen-backbone training windows, then joint. VALID: stratified sample (start → best). Test: stratified estimate, and the full split for the selected cells.

Row	Backbone	Loss	Head	Schedule	VALID start	VALID best	Test (strat.)	Test (full)
shear_flow	MLP neck	vrmse	gate	pre	0.0284	0.0284	0.0280	—
shear_flow	register neck	vrmse	gate	pre	0.0275	0.0275	0.0271	—
shear_flow	3-D register neck	vrmse	gate	pre	0.0307	0.0307	0.0302	0.0303
shear_flow	3-D MLP neck	vrmse	gate	pre	0.0286	0.0286	0.0281	—
shear_flow	vanilla	vrmse	gate	pre	0.0301	0.0294	0.0298	—
rayleigh_benard	MLP neck	vrmse	gate	joint	0.1436	0.0681	0.0659	—
rayleigh_benard	MLP neck	vrmse	gate	pre	0.0454	0.0437	0.0438	—
rayleigh_benard	register neck	mse	gate	joint	0.1426	0.1426	0.1415	—
rayleigh_benard	register neck	vrmse	gate	joint	0.1426	0.0777	0.0742	—
rayleigh_benard	register neck	vrmse	gate	pre	0.0422	0.0422	0.0425	—
rayleigh_benard	register neck	vrmse	plain	joint	0.4470	0.4470	0.4331	—
rayleigh_benard	3-D register neck	vrmse	gate	pre	0.0425	0.0425	0.0429	0.0428
rayleigh_benard	3-D MLP neck	vrmse	gate	pre	0.0423	0.0423	0.0426	0.0426
rayleigh_benard	vanilla	vrmse	gate	joint	0.1441	0.0891	0.0854	—
rayleigh_benard	vanilla	vrmse	gate	pre	0.0438	0.0438	0.0441	—
TRL2D	MLP neck	vrmse	gate	pre	0.1930	0.1918	0.1922	—
TRL2D	register neck	vrmse	gate	pre	0.1928	0.1919	0.1921	—
TRL2D	vanilla	vrmse	gate	pre	0.1926	0.1916	0.1921	—
MHD_64	register neck	vrmse	gate	pre	0.1723	0.1714	0.1728	—
MHD_64	vanilla	vrmse	gate	pre	0.1722	0.1714	0.1727	—
shear_flow	E-ft (3 backbones, equal)	vrmse	gate	pre	—	0.0256	—	0.0252
rayleigh_benard	E-ft (3 backbones, equal)	vrmse	gate	pre	—	0.0400	—	0.0403
TRL2D	E-ft (3 backbones, equal)	vrmse	gate	pre	—	0.1917	—	0.1909
MHD_64	E-ft (2 backbones, equal)	vrmse	gate	pre	—	0.1726	—	0.1725 (vanilla)

Three readings. First, the closed-form gate on the frozen backbone already captures the gain. After exact gate pre-training, the joint stage improves VALID by at most 0.001, and on shear_flow the best checkpoint is the starting one, so end-to-end retraining is not needed. Second, the loss matters only together with the gate. On rayleigh_benard, MSE with the gate gives 0.1415, VRMSE without the gate 0.4331, and both together 0.0425–0.0742. The VRMSE loss gives low-variance onset windows the gradient that MSE takes away; the credit goes to that classical loss, not to the neck. Third, the three backbones tie within one seed (TRL2D 0.1921–0.1922, MHD_64 0.1727–0.1728), and no register edge is claimed from the single-seed differences on rayleigh_benard and shear_flow. The registered equal-weight ensemble of the three backbones (E-ft; registered before any ensemble score) is chosen on VALID over the best single backbone on rayleigh_benard (0.0400 vs 0.0422) and scores 0.0403 on the full test split. The members' residual fields correlate at 0.59–0.64, symmetrically across pairs, so the gain is ordinary variance reduction; the register neck is one member and gets no credit for it. On MHD_64 the two members' residuals correlate at 0.999, the ensemble ties them on VALID, and the selected single backbone scores 0.1725 on the full test split. On shear_flow (residual correlations 0.49–0.68) E-ft is selected and scores 0.0252 on the full test split. On TRL2D it is selected by 0.0001 on VALID and scores 0.1909 on the full test split, below the CNextU-net's 0.1956 and 1% below Flowers' 0.1930 (Muser et al., 2026). There the members' residuals correlate at 0.997–0.999, which bounds the ensembling gain at about 0.0001, so the step from the frozen-backbone gate (0.1921, Table 8) to 0.1909 comes from the gated, metric-aligned fine-tuning itself.

What is left: a transform diagnosis. We decomposed the final models' test error per Fourier octave, per Haar wavelet level and by one-step lag regression (Figure 9; diagnostics only, with no model parameters updated). With $D = y - x_t$ the true change and $P = \hat{y} - x_t$ the predicted change, the magnitude-squared coherence of P and D in an octave gives its Wiener floor, the least error any spatially uniform linear gain on P in that octave can reach. On rayleigh_benard the models sit on this floor in every octave and field; on TRL2D they are within 0.06 of it (in units of the octave's true-change energy); adding the last observed change $x_t - x_{t-1}$ as a second regressor lowers the floor by at most 0.03. Neither the tested scalar gain nor this two-regressor filter removes what is left; longer-lag or cross-field linear filters, and nonlinear, cross-scale or local models, might. In the temporal diagnostic, the model's lag-one regression slope of each octave's next change on its last observed change (teacher-forced, one step; not the model's own pole) matches the true slope within 0.02 on TRL2D and rayleigh_benard; in the top octaves both sit near $-\frac{1}{2}$, the value that frame-to-frame white content gives (differenced i.i.d. noise gives exactly $\mathrm{corr}(D_{t+1}, D_t) = -\frac{1}{2}$, and

approximately so when the signal's own changes are negligible). On shear_flow the models' slope there is near 0 against the truth's -0.43 to -0.51, the one mismatch we found. Wavelets locate the error: on TRL2D 98% of its energy lies in 5% of the Haar coefficients, along the mixing interface, and pressure carries most of it; on rayleigh_benard the worst windows are convection onset (t_0 11–30), where a thin, grid-scale layer at one wall starts to move, 62% of the buoyancy error sits at the finest level, and the octave-4 floor is 0.34–0.50. A deterministic model's least-squares answer to such unpredictable content is to shrink it, which the amplitude ratios (0.72–0.88 in the top octave) show; probabilistic prediction is the principled alternative (Appendix S). A registered two-tap filter per octave on top of the final models ($\alpha P + \beta(x_t - x_{t-1})$, fitted on training windows) improves TRL2D's VALID score by 0.12%, below its 0.5% bar, and is dropped. On shear_flow, where the slope mismatch is, it improves VALID by 16% and the stratified test sample from 0.0252 to 0.0202 (72% of windows lower; scored once). Its fitted coefficients say where the model failed: the large-scale pressure change is better predicted by continuing the last observed change ($\beta = 0.72$) than by the model's own change ($\alpha = 0.31$), and the top pressure octaves fall back to persistence with $\beta \approx -\frac{1}{2}$, the white-content reversion. The gain is metric alignment again: on a subsample of the test windows it comes from the three lower quintiles of target variance ($+29$ to $+33\%$), while the two upper quintiles get worse (-15%, -30%), so the filter corrects quiet windows, not active flow. It is classical and one-step, acts on velocity components separately (incompressibility is not enforced), and its behaviour inside a rollout is untested. Finally, against the released checkpoint on the same 873 TRL2D test windows, the final model is lower on 6 of 9 test trajectories (the other three within 0.0006): paired Wilcoxon over trajectories $p = 0.049$, trajectory-bootstrap 95% CI of the mean difference $[-0.0046, -0.0006]$.

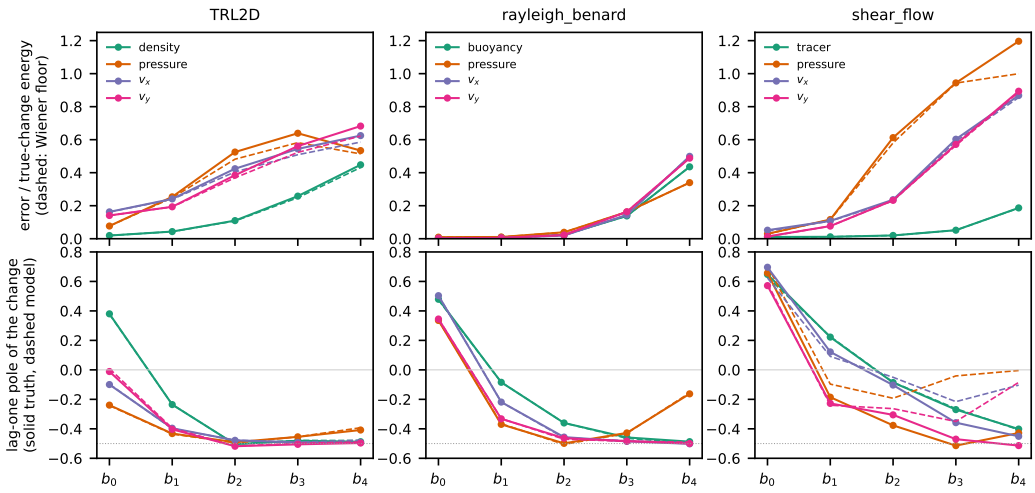


Figure 9: What is left after the gate, per Fourier octave b_0 (coarsest) to b_4. Top: error energy over the octave's true-change energy (solid) and its Wiener floor, $1-$coherence (dashed). Bottom: lag-one regression slope of the octave's next change on the last observed change (teacher-forced), truth (solid) against the final model (dashed); the dotted line marks $-\frac{1}{2}$, the value frame-to-frame white content gives. TRL2D: full test split; rayleigh_benard and shear_flow: stratified test samples.

J Per-seed results for the TRL2D official and sliding-start protocols

Four-seed replication. Registered before any run of seed 1340: one more seed per arm, with the same recipe, checkpoint selection (best rollout VALID) and evaluation. The effect counts as replicated only if seed 1340's primary test is significant against both fine-tuned controls, Holm across four seeds. It is not. At seed 1340 all three arms selected checkpoints from the first 72–84 steps (seeds 1337–1339: steps 5,726–9,309), and the register is the worst of the three: start-averaged 13:30 VRMSE 0.9568 against 0.8996 (MLP) and 0.8939 (vanilla), worse on all nine trajectories against each. Over four seeds the register is significant against the MLP on 2/4 seeds and against

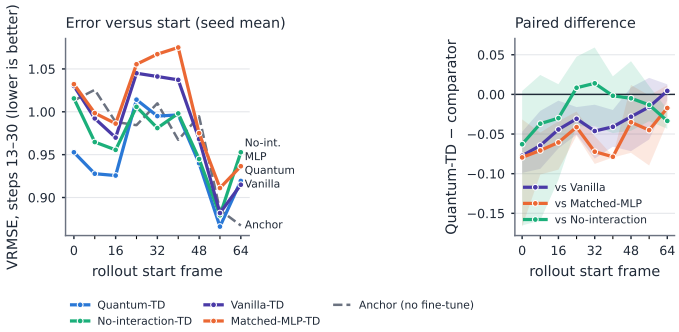


Figure 10: Rollout error versus start frame under the sliding-start protocol. *Left:* seed-mean 13:30 VRMSE per arm at each start (anchor dashed). *Right:* paired difference, register minus comparator (seed mean, band = seed range; below zero = register better). The edge over the fine-tuned classical controls persists across starts; the difference from the anchor is negative only at early starts.

Table 11: TRL2D long-horizon window 13:30 test VRMSE (TD fine-tuning, 3 seeds). Right: seeds on which the register beats that model, by the pre-registered test over start frames (Holm-corrected) and by an uncorrected test over all 81 trajectory-start pairs. Per-seed values and bootstrap intervals: Appendix J.

Model	from the official start frame	mean over 9 start frames	seeds better (Holm)	seeds better (81 pairs)
+ 30-atom register	$\mathbf{0.9528 \pm 0.0542}$	$\mathbf{0.9486 \pm 0.0231}$	—	—
+ register, V_{ij}=0 (no interactions)	1.0156 ± 0.0433	0.9664 ± 0.0264	1/3	1/3
released checkpoint (anchor)	1.0132	0.9707	0/3	1/3
+ register, mean-field (Gutzwiller)	1.0495 ± 0.1219	0.9782 ± 0.0418	0/3	0/3
vanilla continued	1.0301 ± 0.0449	0.9867 ± 0.0196	**2/3**	**3/3**
+ matched MLP	1.0322 ± 0.0259	1.0041 ± 0.0160	**2/3**	**3/3**

vanilla on 1/4 (Table 12). The four-seed start-averaged means are 0.9506 (register), 0.9635 (vanilla) and 0.9780 (MLP), and the per-seed tests support no edge. A secondary analysis scores every arm's final step-10,000 weights with the same tests. It was registered after the early selection was seen for the register and before any final-step evaluation, and it cannot change the primary verdict. There the register is significant against the MLP on 0/4 seeds and against vanilla on 2/4, so the edge over the MLP seen on three seeds depended on validation-based checkpoint selection.

Table 12: TRL2D TD replication, four seeds (1337–1340): 13:30 test VRMSE (mean ± SD over seeds) at the VALID-selected checkpoints (primary) and at the final training step (secondary), the seeds on which the register is significantly better (per-trajectory Wilcoxon on start-averaged error, Holm across four seeds), and the seed-1340 official-start value.

Checkpoint	arm	official start	all 9 starts	reg. better (Holm, 4 seeds)	seed 1340
VALID-selected	register	0.9660 ± 0.0515	0.9506 ± 0.0193	—	1.0056
	matched MLP	1.0101 ± 0.0490	0.9780 ± 0.0539	2/4	0.9439
	vanilla continued	0.9892 ± 0.0897	0.9635 ± 0.0491	1/4	0.8664
final step	register	1.0467 ± 0.0684	1.0242 ± 0.0203	—	1.1009
	matched MLP	1.0109 ± 0.0507	0.9723 ± 0.0373	0/4	0.9841
	vanilla continued	1.1236 ± 0.1669	1.0877 ± 0.0978	2/4	1.1456

Table 13 lists every TD arm's per-seed official-start (start-0) 13:30 test VRMSE, its all-start value, and its rollout-validation selection best (VRMSE@8 at the selected checkpoint). Table 14 gives the pre-specified statistics of the register against each comparator. The sliding-start protocol reproduces the official protocol exactly at start 0 (checked to machine precision). Its primary test is a per-trajectory paired Wilcoxon on 13:30 VRMSE averaged over the nine starts (n=9, one-sided, Holm across seeds); the secondary uses per-(trajectory, start) pairs (n=81, correlated within trajectory,

uncorrected). Per-window p-values on the one-step endpoint are anti-conservative because windows overlap within trajectories and are read together with seed-level consistency.

Arm	13:30, official start (s1337/s1338/s1339)	13:30, all starts	selection best
official checkpoint (anchor)	1.0132	0.9707	—
vanilla continued	1.0787 / 1.0216 / 0.9901	1.0091 / 0.9784 / 0.9727	0.5846 / 0.5909 / 0.5750
+ matched MLP	1.0328 / 1.0060 / 1.0578	0.9993 / 1.0221 / 0.9911	0.5716 / 0.5796 / 0.5802
+ 30q, V_{ij}=0	1.0202 / 0.9701 / 1.0564	0.9437 / 0.9602 / 0.9954	0.5923 / 0.5807 / 0.5739
+ 30q, mean-field	1.1882 / 0.9592 / 1.0011	1.0222 / 0.9732 / 0.9390	0.5847 / 0.5870 / 0.5838
+ 30q Rydberg	0.9927 / 0.9745 / 0.8911	0.9620 / 0.9618 / 0.9219	0.5783 / 0.5771 / 0.5642

Table 13: Per-seed values behind Table 11. The mean-field arms' official-start 6:12 values are 0.5029/0.5221/0.5397; all arms' 6:12 means are 0.5216–0.5278 (anchor 0.5482).

comparator	seed	mean diff	prim. win	prim. p	Holm	sec. win	sec. p	bootstrap CI_{95}
vanilla	1337	-0.0471	0.78	0.0195	0.0391*	0.59	5.0×10^{-3}	$[-0.0833, -0.0129]$
vanilla	1338	-0.0167	0.78	0.1016	0.1016	0.57	2.3×10^{-2}	$[-0.0393, +0.0101]$
vanilla	1339	-0.0508	0.89	0.0039	0.0117*	0.77	7.5×10^{-8}	$[-0.0702, -0.0260]$
matched MLP	1337	-0.0372	0.78	0.1016	0.1016	0.53	1.9×10^{-2}	$[-0.0795, +0.0079]$
matched MLP	1338	-0.0603	1.00	0.0020	0.0059*	0.75	5.2×10^{-8}	$[-0.0889, -0.0339]$
matched MLP	1339	-0.0692	1.00	0.0020	0.0059*	0.77	1.0×10^{-8}	$[-0.0929, -0.0462]$
anchor	1337	-0.0086	0.56	0.5449	1.0000	0.44	4.9×10^{-1}	$[-0.0981, +0.0725]$
anchor	1338	-0.0089	0.44	0.6328	1.0000	0.48	5.0×10^{-1}	$[-0.0900, +0.0571]$
anchor	1339	-0.0488	0.56	0.1504	0.4512	0.56	3.5×10^{-2}	$[-0.1206, +0.0249]$
V_{ij}=0	1337	$+0.0183$	0.11	0.9512	1.0000	0.30	1.0	$[-0.0706, +0.0737]$
V_{ij}=0	1338	$+0.0016$	0.44	0.5000	1.0000	0.54	4.6×10^{-1}	$[-0.0333, +0.0391]$
V_{ij}=0	1339	-0.0735	0.89	0.0039	0.0117*	0.67	9.5×10^{-5}	$[-0.1152, -0.0350]$
mean-field	1337	-0.0602	0.44	0.4102	0.5703	0.40	5.4×10^{-1}	$[-0.2393, +0.0877]$
mean-field	1338	-0.0115	0.44	0.2852	0.5703	0.53	2.0×10^{-1}	$[-0.0349, +0.0090]$
mean-field	1339	-0.0172	0.56	0.1504	0.4512	0.51	1.2×10^{-1}	$[-0.0447, +0.0101]$

Table 14: Sliding-start protocol: register minus comparator on start-averaged 13:30 VRMSE, per seed. *: Holm-adjusted $p < 0.05$. CI: trajectory cluster bootstrap (10,000 resamples) of the mean paired difference.

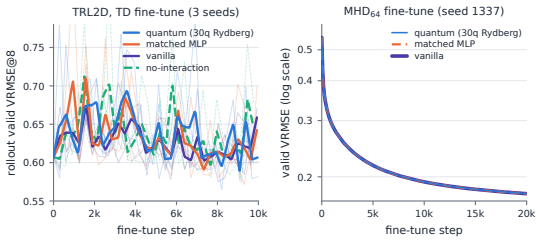


Figure 11: Fine-tuning dynamics. *Left:* TRL2D TD fine-tuning from the official checkpoint (thin: seeds; bold: 3-seed mean of the rollout-validation selection metric); arms overlap step by step, and the test-rollout separation is not visible in validation curves. *Right:* `MHD_64` (seed 1337, log scale); all arms converge to per-channel validation VRMSE ≈0.177.

K QUBIT-LADDER DETAILS

Per-seed VRMSE@8 (s1337/s1338/s1339) under TD: 5 atoms 0.9245/0.8053/0.7982; 10 atoms 1.1810/0.7799/0.8387; 20 atoms 1.2486/0.7843/0.7478; 30 atoms 0.903/0.826/0.806; matched MLP 1.098/0.933/0.829; mean-field pilot 0.891/0.825/0.802. Seed 1337 is the outlier at every interior count. Under the one-step arena objective the means were 30 atoms 1.1002 ± 0.1939, matched MLP

1.0574 ± 0.0574 and bare backbone 0.9637 ± 0.0048: the adapters' edge appears only under TD. A smoke-scale CMA-ES configuration search (about 200 evaluations per regime) was not competitive with gradient training.

L MHD_64

The released `MHD_64` checkpoint evaluates to 0.5447 (test) / 0.5464 (validation) per-channel VRMSE under our pipeline (≈0.66 field-grouped, the convention of the published 0.1633); the same pipeline reproduces the TRL2D anchor to 1.1%, which points to, but does not prove, a discrepancy in the released artifact. Four of the nine 20k-step runs stopped at about 7.5k steps; the primary row uses the two seeds at which all three arms reached the same length (backbone 0.1897 ± 0.0178, MLP 0.1897 ± 0.0178, register 0.1896 ± 0.0177; the arms agree to four decimals and the gate never opens). All arms beat the released artifact on every one of 960 test windows.

Physics constraints on the classical part (registered). The true magnetic field is divergence-free on this periodic grid (longitudinal energy share 8×10^{-6}), while 6% of the backbone's field error is longitudinal, and the backbone, which zero-pads and down-samples, is not shift-equivariant. Projecting the predicted field onto its divergence-free part in physical units lowers the gated model's full-test VRMSE from 0.1725 to 0.1709 per channel (0.1853 to 0.1840 field-grouped); averaging over eight circular shifts of the input as well gives 0.1592† (0.1707†), lower in every channel and in every octave, with the predicted high-wavenumber energy shrinking by about 3%. †Eight-fold test-time averaging, eight times the inference cost; the published model did not average. On the published convention both remain above 0.1633 and far above the current best, 0.1160 (EddyFormer, Du & Krishnapriyan, 2025; Flowers 0.1165). Both steps change the classical part of every arm equally.

M LEARNED REGISTER CONFIGURATIONS

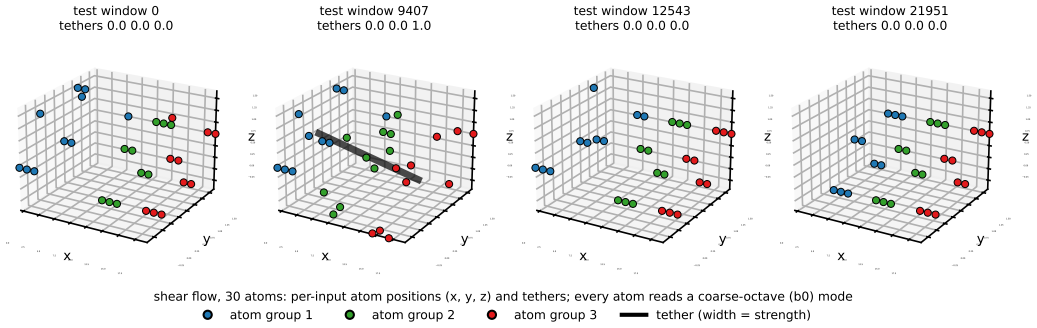


Figure 12: Per-input qubit configuration in 3-D (routed register, 3-D variant of Appendix P, shear flow, 30 atoms, seed 1): atom positions (x, y, z) of the three 10-atom groups (drawn side by side) on four test windows, and tether strength between groups (line width). Positions move with each input (mean 0.33, at most 1.36 grid units from the mean layout; z spans 2.1), and on window 9407 the network switches on the tether between groups 1 and 3. The variation is shown, not shown useful: swapping positions or tethers between inputs changed VALID error by at most 0.21% (registered, 3 seeds).

N VISUAL ROLLOUT FIELDS: TRL2D AND MHD_64

O DYNAMIC CONFIGURATION AND HARDWARE MAPPING

Table 15 states what a deployment would need. The neck register's 2-D geometry and global drives map directly, and distances are clamped from below as both devices require. Three features do not map as trained. Signed, per-layer, input-dependent local detunings would have to be re-encoded

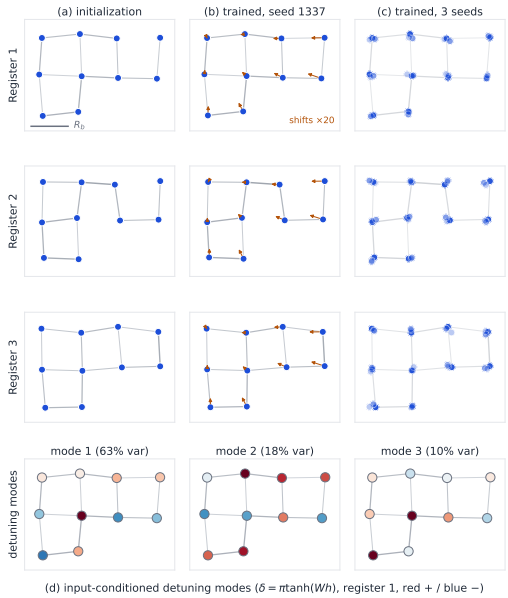


Figure 13: Register configurations (three blockade-decoupled 10-atom registers, trainable 2-D geometry, blockade-radius units). *(a)* Initialization (lattice with σ=0.05 R_b jitter). *(b)* TD-trained geometry (seed 1337, displacements $\times$20): a small coherent contraction (mean radial shift $-0.007\,R_b$). *(c)* Three seeds agree to $\leq 0.11\,R_b$ rms. *(d)* Per-input reconfiguration via detunings: the top three singular modes of the conditioning map hold 91% of its variance.

Table 15: What maps to current neutral-atom devices, from the vendors' public programming interfaces (QuEra Aquila via Amazon Braket's analog Hamiltonian simulation schema; Pasqal Fresnel and Pulser's detuning map modulator, DMM).

	Our registers	QuEra Aquila	Pasqal Fresnel
Geometry	2-D, 3 × 10 atoms; routed co-processor adds per-input 2-D offsets	user-defined 2-D arrangement	2-D on calibrated trap layouts; ≤80 atoms, ≥5 µm apart, within 38 µm
Drive	global Ω, ϕ per layer	global $\Omega(t)$, $\phi(t)$	global Rydberg channel
Detuning	signed, per atom, per layer, input-dependent	global $\Delta(t)$; local $\Delta_\ell(t)\,h_k$, $h_k \in [0,1]$ static (experimental)	global $\Delta(t)$; DMM: negative only, one waveform, static weights in $[0,1]$
Readout	$\langle Z\rangle$, $\langle X\rangle$, $\langle Y\rangle$	ground/Rydberg bitstrings	ground/Rydberg bitstrings

as a global detuning plus one static, one-signed weight pattern per input, which means one device program per input. $\langle X\rangle$ and $\langle Y\rangle$ readouts need a basis-rotating pulse before measurement. The configuration network also outputs a z offset, but it stays at zero in training (a planar start gives it no gradient), so every learned layout is 2-D.

Finite-shot readout. A device estimates each atom's $\langle Z\rangle$, $\langle X\rangle$ and $\langle Y\rangle$ from N shots per measurement setting. We re-evaluated the route-supervised co-processor register (3 seeds per row, trained exactly as before) with each expectation replaced by $2\,\mathrm{Bin}(N,(1+\langle P\rangle)/2)/N-1$, drawn independently per atom and axis. This independent-marginal approximation gets each estimate's variance right but omits the covariance between atoms read in the same shot (a real shot samples a joint bitstring), so it is not a hardware-fidelity test. Nothing was retrained with noise, and the rule was registered first: under this approximation the register counts as shot-robust at N if the mean increase in test VRMSE stays below half its gap to the matched MLP. The increase is 7.6×10^{-5}, 8×10^{-6} and 1×10^{-6} on shear_flow and 3.2×10^{-4}, 3.4×10^{-5} and 4×10^{-6} on rayleigh_benard at $N=100$, 1,000 and 10,000 (thresholds 5.5×10^{-5} and 1.8×10^{-4}). The register is shot-robust under this approximation from $N=1{,}000$ on both rows and not at $N=100$. Because that gap is itself not significant, we also compare with seed spread: at $N=1{,}000$ the increase is below the

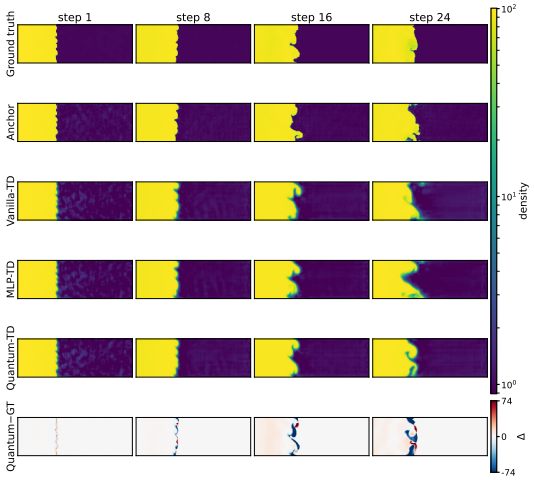


Figure 14: Ground truth vs model rollouts on the first TRL2D test trajectory (density, log scale; bottom: signed error of the TD register arm). Error concentrates at the mixing front. On this trajectory the anchor has the lowest late-step VRMSE (0.640/0.659 at steps 16/24 vs 0.890/0.712 for the register arm); headline claims rest on full-test-set averages.

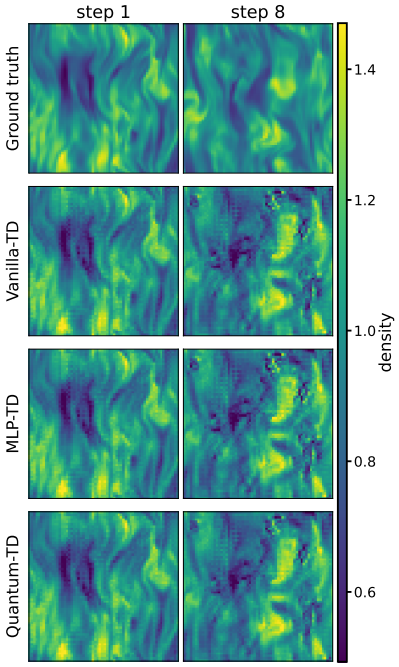


Figure 15: Ground truth vs model rollouts for 3D magnetohydrodynamics (MHD_64, valid trajectory 0, mid-z density slice, steps 1 and 8). Top to bottom: Ground truth, Vanilla-TD, matched MLP-TD, and Quantum-TD. All arms preserve coarse density topology; fine discrepancies concentrate in turbulent magnetic shear interfaces.

seed standard deviation of every core on shear_flow (1.8–3.4×10^{-4}) and of the classical cores on rayleigh_benard (3.1–3.7×10^{-4}), though not below the register's own spread there (2.2×10^{-5}). With the three 10-atom groups run in parallel on a 30-atom array, the measured cost per input is $3N$ shots (three settings) when the tethers are inactive, as in five of the six shot-study fits (their tethers are fixed, not set per input); the single nonzero tether (shear_flow seed 1, group 0 to group 2) needs two rounds, $6N$, and a full three-group chain would need up to $9N$. A 10-atom device reused in turn needs $9N$ group executions. The matched classical cores need none of this. Calibration errors were tested the same way (registered first; 3 seeds, 3 repeats): Gaussian in-plane position jitter of 1% and 5% of the lattice spacing and a multiplicative detuning error of 1% and 5% per atom and layer. Jitter and the 1% detuning error raise test VRMSE by at most 7×10^{-6} on both rows; a 5% detuning error costs 1.4×10^{-4} on rayleigh_benard (within its threshold) and 7.3×10^{-5} on shear_flow (above its 5.5×10^{-5} threshold). The register tolerates placement errors well and needs detunings calibrated to about 1%.

Detunings depend on each input; geometry, drives and evolution times are shared learned parameters in the reported arms, and each forward pass resets the register. An arena-only variant with input-conditioned positions $r_i(h) = r_i^0 + 0.6 \tanh(g_\phi(h)_i)$ reached 0.8951 ± 0.1367, worse than the static layout. Mapping to a neutral-atom device would require: replacing signed per-layer local detunings with the platform's addressable detuning patterns; enforcing minimum atom separation (clamping regularizes interaction energies but does not enforce physical spacing); replacing the first-order digital–analog split by continuous pulses; and a noise model beyond finite-shot readout. Optical tweezers assemble user-defined 2-D arrangements at the 100-atom scale (Barredo et al., 2018), with coherent arrays reported at 6,100 atoms (Manetsch et al., 2025); the registers here use at most 30 atoms in blocks of 10. The trained neck registers of the audit are inference-inert, so

those deployed models need none of this; the routed co-processor is not, and its readout cost is given above.

P HYBRID SCALING WITH THE QUBIT BUDGET

Table 1's convection row was first scored on a stratified test sample (5,600 of 34,300 windows), the sample this scaling study uses: classical 0.0449, +MLP 0.0428, +30/50/100 qubits 0.0426/0.0422/0.0412. A registered retraining with the same recipe, scored once on every test window, gives the values now in the table (0.0456, 0.0435, 0.0430/0.0425/0.0418; vs MLP $+1.1\%$ instead of $+0.5\%$, vs classical $+5.6\%$ instead of $+5.1\%$). It is a representativeness check on test data already used, not an independent confirmation.

Registered before the sweep (a 5-qubit smoke run only tested the code). K counts allocated atom slots across groups of at most 10 atoms: the largest joint quantum state has 10 atoms, and tethers pass classical readouts between groups, so 200 qubits means 20 separately simulated 10-atom groups, not a 200-qubit entangled state. The backbone and cached inputs are fixed; the gate, a classical head for unselected modes and a classical bypass beside the core on selected modes are retrained with the core at each budget and seed, so budgets compare separately fitted hybrids. The routed co-processor sets every register degree of freedom it can per input: atom positions (x, y, z), detunings, which modes the atoms receive, and, new here, the strength of each tether between the 10-atom groups (a learned base plus a linear map of the routed modes' features). Drives and evolution times stay shared. Budgets $K \in \{0, 5, 10, 20, 30\}$ ($K = 0$: classical part alone); register and a similarly sized MLP (199 trainable core parameters against the register's 241, 244 in total); three seeds on every row (full test on shear_flow, TRL2D and MHD_64, stratified test on rayleigh_benard). The backbone in the classical part is the vanilla CNextU-net, never trained with a register. A first version of this sweep used, on three rows, a backbone trained with a register in its neck; it is kept below as a check of how much the backbone matters.

Figure 5 (top) gives the hybrid's test VRMSE relative to its classical part. It falls with the budget on the two rows with predictable error: to 0.877 (register) and 0.900 (MLP) at 30 qubits on shear_flow, and to 0.949 and 0.954 on rayleigh_benard. It stays at 1.000 on TRL2D and MHD_64. Figure 16 shows where the added qubits work. The bottom row of Figure 5 explains the shape in error-energy space, where orthonormal modes add. The coverage $U(K)$, the best K modes' share of the error energy, is measured, like the captured share, relative to the error energy of the cached pre-gate anchor, not of the learned gate's residual, and is computed from the test data for any K (no fitting); each run's own coverage bounds what its K qubits can capture, while the plotted curve pools runs and is descriptive. It follows the closed form $U_\infty[1 - (1 + K/K_0)^{1-a}]$ within 0.015 on every row (0.008 or less except on TRL2D; U_∞ 0.430, 0.0475, 0.555, 0.019 and a 1.92, 2.11, 1.85, 2.62 for shear_flow, rayleigh_benard, TRL2D, MHD_64), a power-law decay of mode energies. $U(K)$ saturates near 200 qubits because the band-wise POD basis spans only that share of the anchor's error energy: the limit beyond it is the basis, not the qubit count, and a richer or input-adaptive basis (wavelet packets, a best basis per input) is what would raise it. The qubits reach $\eta(K) =$ mean captured/mean $U(K)$ of the coverage, where $U(K)$ is the curve pooled over both cores, all budgets and seeds (each arm's gate is trained separately, so a run's own coverage differs; against the register runs' own coverage η is 0.78 and 0.81 on shear_flow and 0.81 and 0.83 on rayleigh_benard at 100 and 200 qubits), rising from 0.34 to 0.63 on shear_flow and from 0.42 to 0.65 on rayleigh_benard between 5 and 30 qubits; on TRL2D U_∞ is large but η is effectively zero, the residual no tested corrector captured (Appendix I). At 30 qubits the register captures 54% of what the basis allows on shear_flow (86% at its current efficiency); the shaded fans extrapolate beyond 30 qubits with η held at its 30-qubit value or at 1. The fan was tested at 50 qubits (five groups, same per-input layout), with ranges for the register's capture registered before any run: $[0.249, 0.380]$ on shear_flow and $[0.027, 0.040]$ on rayleigh_benard. The upper edges come from the pooled coverage (only a run's own coverage is a strict bound on its capture), so the test is whether η falls below its 30-qubit value. All six seeds fell inside (shear_flow 0.255, 0.253, 0.269; rayleigh_benard 0.029, 0.028, 0.028), near the lower edges, with η rising slightly (0.63 to 0.66 and 0.65 to 0.70), and the hybrid reached 0.869 and 0.940 of its classical part. The matched MLP did as well or better (shear_flow 0.268, 0.263, 0.302; rayleigh_benard 0.028, 0.027, 0.028; hybrid 0.877 and 0.940), so at 50 qubits too the gain belongs to the routing, not the register. At 100 qubits a registered replication on five new seeds (4–8), with the rule fixed before any 200-qubit register result, found the one exception: on shear

flow the register's test VRMSE was below its matched MLP's on 5 of 5 seeds (means 0.0202 against 0.0207, 2.5%; one-sided paired t-test $p = 0.0065$, below 0.05 even after a Holm correction over the four registered cells). This MLP has 199 trainable core parameters against the register's 241 (244 in total; the MLP's hidden width is an integer), and it does not see the register's interaction graph; a one-unit wider MLP (297) and a graph-aware GNN of 239 parameters that learns its own interaction graph with the same geometry architecture, both registered before they ran, were scored on all five shared seeds. The register was lower than the GNN on only 2 of the 5 seeds (means 0.02020 against 0.01990; one-sided paired $p = 0.76$) and lower than the wider MLP on 3 of 5 (0.02020 against 0.02053, $p = 0.13$), so it does not meet the rule (lower on at least 4 of 5 and $p < 0.05$) against either: the edge is not established against a graph-aware GNN of similar size or a wider MLP. The GNN learns its own graph and router and normalises its coupling weights, so this does not attribute the original edge to the graph. The lower mean VRMSE is not accompanied by a consistently higher captured-energy share (the register's is lower on 3 of the 5 new seeds); the two measures weight fields and windows differently, and the co-trained gate and head differ between arms. A metric-aligned 100-qubit control remains pending; the test windows were seen in earlier rounds, so this is TEST-informed. On rayleigh_benard the register is below the smaller MLP on 5 of 5 new seeds (0.0410 against 0.0414, 0.9%, one-sided paired $p = 0.029$), which passes the unadjusted rule; the Holm correction over the four cells waits for the two 200-qubit cells. Against the same two controls on this row the register was lower than the GNN on 1 of 5 seeds (0.04104 against 0.04078, $p = 0.97$) and than the wider MLP on 3 of 5 (0.04104 against 0.04114, $p = 0.15$), so it does not meet the rule against either. A registered interaction-free control (the same register retrained with interaction time fixed at zero, so each atom evolves alone; 3 seeds, VALID) is within 0.5% of the register on all 3 seeds on rayleigh_benard (no measurable contribution from atom–atom interaction in this recipe) and mixed on shear_flow ($+1.3$, $+0.2$, -0.7%; neither outcome of the rule). Training on the metric itself (mean VRMSE), promoted on VALID by a registered rule and then scored once on TEST from frozen checkpoints (TEST-informed, two seeds), lowered TEST error for the register (shear_flow 0.0195 and 0.0199 from 0.0205 and 0.0208; rayleigh_benard 0.0418 and 0.0417 from 0.0425 and 0.0427) and by similar amounts for the matched MLP. The metric-aligned loss lowers TEST error for both cores in all four row-seed pairs, without establishing a register-specific advantage. About 94% of the capture added between 30 and 50 qubits lands in the coarsest octave (shear_flow 0.222 to 0.248 of the error, rayleigh_benard 0.020 to 0.025); the finer octaves, whose error no tested corrector predicts (Appendix I), barely move. The same ladder at 50, 100 and 200 qubits on TRL2D and MHD_64 (registered before any run with the expectation that η stays below 0.15 at every budget; the stated rationale was that their gated residual is not predictable from the input) leaves the hybrid on its classical part: seed-mean test VRMSE 0.1922 against 0.1922 on TRL2D and 0.1733 against 0.1733 on MHD_64 at every budget (single seeds differ in the fourth decimal), for the register and the matched MLP alike (three seeds each). On TRL2D the qubits capture effectively nothing ($\eta \leq 2.4\times10^{-4}$; at most 1.2×10^{-4} of the error, on one seed). On MHD_64 the register's capture rises from 0.0015 to 0.0020 of the error (seed means at 50 and 200 qubits; η 0.10 to 0.11), inside the registered ranges on all nine seeds, and the matched MLP captures as much (0.0021 at 200). More qubits did not recover the remaining error with this basis, router and trained correction; whether another input-conditioned correction could is not tested here.

100 and 200 qubits (registered). The same test at 100 and 200 qubits (10 and 20 groups), with ranges registered before any run from η at 50 qubits: $[0.274, 0.405]$ and $[0.279, 0.412]$ on shear_flow, $[0.0313, 0.0442]$ and $[0.0323, 0.0455]$ on rayleigh_benard. All twelve seeds fell inside (shear_flow 0.348–0.354 and 0.369–0.375; rayleigh_benard 0.037–0.039 and 0.040–0.041), η rose to 0.81 and 0.84 at 100 qubits and 0.88 and 0.87 at 200, and the hybrid reached 0.837 and 0.917 of its classical part at 100 qubits and 0.851 and 0.919 at 200. Past 100 qubits the coverage saturates (U 0.416 at 100 and 0.423 at 200 on shear_flow), so the extra qubits have little left to take. The matched MLP reached 0.891 and 0.923 at 100 and 0.859 and 0.917 at 200; the register is below it on shear_flow on 3/3 seeds at 100 qubits (paired $p = 0.006$) and 2/3 at 200. That comparison was not the registered test; a registered replication on five new seeds decides it.

Where the gain comes from (registered, 3 seeds). The position head starts at zero and every atom at z=0, so the loss has no gradient in z and all trained layouts are 2-D. A control registered before it ran compares, at 30 atoms with the same routing design (each model trained separately), per-input 2-D positions with dynamic tethers (shear_flow 0.02073, rayleigh_benard 0.04261), a fixed grid without tethers (0.02071, 0.04258), and a bilayer grid that trains z (0.02110, 0.04271);

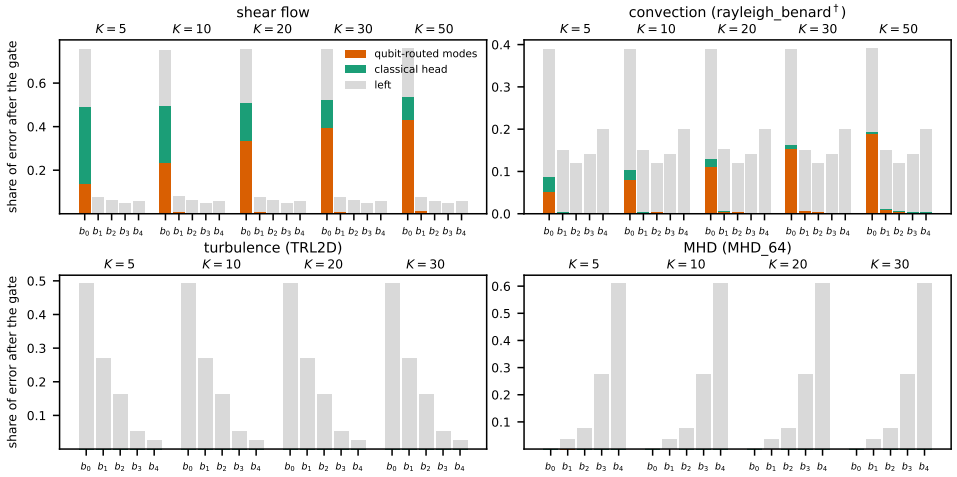


Figure 16: Where added qubits work, per Fourier octave (b_0 coarsest to b_4 finest), register arm, three seeds. Bars (fractions of the pre-gate anchor's error energy): error after the gate captured by the qubit-routed modes (orange), by the classical head (green), and left (grey). On shear flow and convection the error sits in the coarse octave and the qubits take it over from the classical head as K grows; on TRL2D it is spread across octaves and on MHD it sits in the finest octave, and no budget up to 200 qubits captures it here.

the matched MLP scores 0.02129 and 0.04281. Neither "3-D helps" (1/3 seeds on each row) nor "geometry and tethers help" (2/3 and 0/3) passes its rule: no tested geometry or tether variant met the registered gain rule (not an equivalence test). In a validation-only mechanism check registered before its six trainings, swapping positions or tethers across inputs, or replacing positions with the fixed grid, changed the register's VRMSE by at most 0.21% in any seed; none met the prespecified criterion of a $\geq 0.5\%$ degradation on at least two of three seeds per row. Removing the fitted core term at inference raised VRMSE by 4.8–17.4% on shear_flow and 14.2–19.8% on rayleigh_benard. Thus the fitted core term contributes at these checkpoints, while the tested per-input configuration perturbations have not shown a material effect under the registered rule.

3-D, tethered configuration on every benchmark (registered). A 3-D register (three 10-atom groups; per-input positions (x, y, z) on a randomly offset 3-D lattice so every coordinate trains; per-input detunings; per-input tethers that pass each group's readout to the next) replaced the neck register on every benchmark, each against its own similarly sized MLP. It improves on the classical model on the heat plume (MAE 0.0341 against 0.0364) and TORAX (VRMSE@8 0.0305 against 0.0321), but its matched MLP gains as much (0.0340, 0.0309); on RealPDEBench every corrector scores 0.00939; on fusion equilibrium it ties its MLP with the starter U-Net (S 0.786 against 0.791), and on the stronger PCA backbone (held-out S 0.9844) it beats its MLP on 3/3 seeds (0.9861 against 0.9849; paired $p = 0.039$ one-sided, 0.078 two-sided), all of it in the ψ-derived shape scalars (elongation R^2 0.900 against 0.892); on the FNO arena it beats its MLP on 3/3 seeds (0.969 against 1.023, p=0.010), which does not survive correction across benchmarks. A registered screen of seven configuration levers in the routed co-processor (3-D random lattice, wider per-input shifts, active tethers, depth, configuration width, their combination, a neighbour-correlator readout) found none that lowers validation error on both rows. Figure 17 shows where the three models differ on one sample per benchmark.

With the register-trained backbone the curves have the same shape: 0.858 (register) and 0.885 (MLP) at 30 qubits on shear_flow, 0.961 and 0.957 on rayleigh_benard, flat on TRL2D.

Exploratory, not the registered test: the register is below its matched MLP in 10 of 12 seed-by-budget cells on shear_flow (mean 1.8%; at 30 qubits 2 of 3 seeds, paired $p = 0.30$) and in 11 of 12 on rayleigh_benard (mean 0.5%; 3 of 3 at 30 qubits, $p = 0.19$); TRL2D and MHD_64 tie. With fixed tether strengths at 30 qubits the two tied (Appendix I). The registered rule requires a GNN arm and a majority of rows, so no quantum claim follows.

Q A DISTRIBUTION OF CONFIGURATIONS PER INPUT

On TRL2D the qubits capture no predictable error at any budget ($\eta = 0$, Appendix P), and the change that the backbone misses behaves like frame-to-frame noise (Appendix I). A single configuration per input can only predict the mean of that noise. WeatherNext 3 (Rasp et al., 2026) builds its ensembles on functional generative networks (Alet et al., 2025): each member is drawn by feeding one low-dimensional noise vector into the network's conditional normalization layers, and training minimizes the fair CRPS over two members. We move the noise into the qubit configuration.

Method. For each window and ensemble member, a code $\xi \sim \mathcal{N}(0, I_{32})$ enters only the input-dependent configuration of Appendix P (30 qubits, band gate, supervised router, per-input tethers). It enters in three places:

- a feature-wise affine map (scale and shift) of the features that enter the core, the analogue of conditional normalization;
- the hidden layer of the network that sets each atom's x, y, z offset;
- the per-input tether logits.

Each input therefore gets a distribution over atom positions, detunings and tether strengths, and each draw is one ensemble member. The three maps start at zero, so training starts at the deterministic hybrid. The gate, the classical head and the routing stay deterministic, so any spread comes through the co-processor alone.

Loss. We minimize the fair energy score over two members (Gneiting & Raftery, 2007). Dividing by the target's own spatial variance $D(y)$, as below, weights each window by $1/\sqrt{D(y)}$, so the score is observation-weighted and not strictly proper (it is minimized by a tilted law); the registered re-run replaces $D(y)$ with a fixed training-set scale, which keeps it proper. In the orthonormal band-wise POD basis it is exact and cheap. Per field, a member's distance to the target is $\sqrt{E_{\text{new},m}/D}$, which is that member's VRMSE. The distance between two members is $\|c_m - c_{m'}\|/\sqrt{D}$. For a point forecast the score equals the VRMSE, so the two share one scale. Unlike a per-point CRPS, which WeatherNext 3 notes can reward members with wrong spatial covariance, it depends on the joint distribution of each field, although its sensitivity to errors in that covariance is known to be weak (Pinson & Tastu, 2013).

Protocol. Selection is on the VALID energy score with 8 fixed draws; TEST is scored once with 16 draws. The matched MLP twin shares every noise path; as in Appendix P, only the register reads the atom positions. We report:

- the energy score;
- the benchmark VRMSE of the ensemble mean;
- the spread–skill ratio (with the $(M+1)/M$ correction);
- the deterministic hybrid's VRMSE at the same seeds, for comparison.

The protocol was fixed before any run, though after earlier TEST results on these rows (TEST-informed, Appendix H). Repeat runs of the deterministic hybrid at one seed differ by up to 3% on the GPU, so single-seed gaps below that are not read.

Results. Table 16 gives the vanilla-backbone rows. On every row and core the energy score is 16–20% below the deterministic hybrid's VRMSE, while the ensemble mean's VRMSE is 1–9% worse. The benchmark metric therefore does not improve, and this lower score is variance-normalised (not strictly proper). The registered proper re-run (fixed training-set scale; shear_flow and rayleigh_benard, seeds 1–3, TEST once) confirms the ordering on every seed: the proper-trained hybrid scores 0.0062–0.0063 and 0.0225–0.0231, against 0.0079–0.0080 and 0.0277–0.0281 for the deterministic hybrid and 0.0069–0.0070 and 0.0234–0.0237 for a Gaussian dressing tuned on VALID under the same score; the matched MLP core scores the same within noise (0.0062–0.0064; 0.0232–0.0235), so the effect is not register-specific. Below, on the variance-normalised score, a

Gaussian dressing beats the drawn configurations on some controls. The spread is calibrated on shear_flow (spread–skill 1.03–1.07) and under-dispersed on TRL2D (0.57–0.58), where it can only come through the 30 routed modes.

Much of the gain comes from calibrated spread, and on TRL2D a control without training does better. A stationary Gaussian dressing of the deterministic hybrid, tuned on VALID, reaches 0.163 (0.175 on the routed modes only). The registered primary control sizes the noise on each mode by that mode's TRAIN residual relative to the window's input amplitude (the standard deviation of the last input frame), times one scale chosen on VALID. On TRL2D it reaches 0.147 with either core, below the drawn configurations even when their spread is scaled about the ensemble mean by a VALID-tuned factor τ (0.152 register, 0.158 MLP, $\tau = 2.5$ on every seed). That τ overshoots the spread–skill ratio (1.2–1.5) and still lowers the score, which suggests the drawn spread has the wrong profile across windows rather than the wrong size. On shear_flow the order reverses on every seed and core (0.017 and 0.018 drawn, 0.019 dressed), as it does on rayleigh_benard (0.035 drawn, 0.037 dressed), while on MHD_64 the dressing wins again (0.128 against 0.139–0.142). The dressing thus wins on the two rows whose remaining error the qubits cannot capture ($\eta \approx 0$, Appendix P) and loses on the two where they capture most of what the basis allows, a pattern over four rows only. Among drawn configurations the register was below its matched MLP on each of the first three seeds on TRL2D and shear_flow (paired $p = 0.055$ and 0.069). An extension to six seeds, registered after seeing those three, decides this by a two-sided paired t test with Holm correction across the two rows and requires the register lower on at least five seeds. On shear_flow it is lower on 6/6 seeds, by 3.4% (Holm $p = 0.044$), which meets the rule (0.0171, against 0.0177 for the MLP, 0.019 dressed and 0.021 deterministic). On TRL2D it is lower on 5/6 seeds by 0.8% ($p = 0.12$), which does not meet the rule, and the untrained dressing beats both cores there. The matched MLP, however, has 199 trainable parameters to the register's 244, so a second control, registered before it ran, widened the MLP by one hidden unit (297 parameters). Against it the register is lower on only 4/6 seeds on shear_flow (2.2%, $p = 0.24$), which fails the same rule: the shear advantage does not survive a wider MLP, and no core effect is claimed. The two MLP widths differ by no more than seed noise (3/6, $p = 0.48$). On TRL2D, reported without a claim, the register is lower than the wider MLP on 5/6 seeds (1.1%, $p = 0.09$).

Table 16: Distribution of configurations per input (30 qubits, TEST, mean over seeds). ES / det.: energy score over the deterministic hybrid's VRMSE, same core and seeds. Last two columns (seeds in brackets): the energy score with the spread scaled by the VALID-tuned τ, and the window-scaled Gaussian dressing of the deterministic hybrid.

Row	core	seeds	deterministic VRMSE	energy score	ES / det.	ensemble-mean VRMSE	spread–skill	ES, tuned τ	dressed (e)
TRL2D	register	3	0.1922	0.1593	0.829	0.1941	0.58	0.1517 (3)	0.1469 (3)
	MLP	3	0.1922	0.1612	0.839	0.1944	0.57	0.1578 (3)	0.1470 (3)
	register vs MLP on the energy score: +1.2% (3/3 seeds, paired $p = 0.06$); tuned τ: +3.9% (3/3, $p = 0.08$)								
shear_flow	register	3	0.0207	0.0170	0.822	0.0226	1.03	0.0171 (3)	0.0189 (3)
	MLP	3	0.0213	0.0179	0.839	0.0231	1.07	0.0179 (3)	0.0190 (3)
	register vs MLP on the energy score: +4.6% (3/3 seeds, paired $p = 0.07$); tuned τ: +4.5% (3/3, $p = 0.14$)								
rayleigh_benard†	register	3	0.0426	0.0352	0.827	0.0444	0.77	0.0350 (3)	0.0368 (3)
	MLP	3	0.0428	0.0356	0.831	0.0447	0.88	0.0358 (3)	0.0369 (3)
	register vs MLP on the energy score: +1.0% (2/3 seeds, paired $p = 0.36$); tuned τ: +2.3% (3/3, $p = 0.06$)								
MHD_64	register	3	0.1733	0.1419	0.818	0.1761	0.63	0.1419 (3)	0.1280 (3)
	MLP	3	0.1733	0.1392	0.803	0.1769	0.79	0.1387 (3)	0.1281 (3)
	register vs MLP on the energy score: -1.9% (1/3 seeds, paired $p = 0.34$); tuned τ: -2.3% (1/3, $p = 0.33$)								

R VEHICLE ROUTING: THE ROUTED REGISTER ON ROUTEFINDER

Row. RouteFinder (Berto et al., 2025) is a pretrained transformer for vehicle routing. Its released RF-TE checkpoint reproduces on the 16 variants with 100 customers to a mean gap of 3.126% (published 3.124%). The row we use is its efficient-adapter-layer (EAL) fine-tuning to 16 unseen multi-depot variants, with their protocol:

- 10 epochs of 10,000 generated instances, batch 128, learning rate 3×10^{-4}, last epoch kept;
- their test files and their gap to reference solutions;
- fp32 for every arm of the qubit ladder; the later 3-D tethered triplet (vanilla, register, matched MLP) ran in bf16 mixed precision, matched within the triplet but not with the ladder.

The fixed classical part ($K = 0$) is the pretrained model fine-tuned this way. A second routing anchor reproduces as well: the released RRNCO checkpoint (Son et al., 2026), which routes on real road-network distance matrices, scores 72.160, 23.279 and 73.005 on its three asymmetric CVRP test sets (in distribution, clustered, out of distribution; published 72.145, 23.282 and 73.010). With the released model frozen and only the routed register trained (RRNCO's own REINFORCE and real-city generator, 10 epochs $\times$ 10k instances, three seeds), the hybrid is worse on all three sets: 3.71, 3.90 and 3.44% against 3.49, 3.57 and 3.22% (gaps against the reference costs implied by the published numbers); after one epoch at a tenth of the rate it sits at parity (3.48, 3.55, 3.20%), so the loss grows with training. RRNCO's own PyVRP local search applied at test time to the released model's best decode lowers the gaps to 2.41, 2.61 and 2.07% (RADAR: 2.61, 2.62, 2.59%), at about 0.09 s of CPU per instance; this is extra test-time compute, not a like-for-like comparison with RADAR's decode-only numbers.

Routed register. A pooled insertion would see only the mean node embedding (Proposition 1). The routed register instead acts per instance:

- a learned scorer picks K nodes;
- their atoms sit at those customers' own coordinates, scaled to lattice units, plus a bounded learned offset in x, y, z;
- groups of at most 10 atoms are joined by tethers whose strengths are set per instance;
- each atom's readout ($\langle Z\rangle, \langle X\rangle, \langle Y\rangle$) is written back to its own node through a zero-initialized map, so training starts at the pretrained model.

The MLP control uses the same routing design, tethers and write-back, trained separately; its core ignores the atom coordinates.

Two fixes before the reported runs. First, the multi-depot environment can reach a state with no feasible action. When a remaining customer cannot be served from any depot, each depot deadlock marks the current depot unavailable until none is left, and sampling then fails. We let such rows visit any unvisited customer. The change acts only where the original code would stop, so it leaves every other step as it was. Second, every arm trains on the same generated instances, because instance differences alone moved one pilot arm by 1.6 points.

Results. Table 17 and Figure 18 give the qubit ladder at two budgets. At the pilot budget the register is ahead of vanilla at every budget tried, and ahead of its matched MLP at 5, 10 and 30 qubits but behind it at 20. At the EAL budget no routed arm improves on the fine-tuned model alone: vanilla 8.53%, matched MLP 8.56% and register 8.59% at 30 qubits; at 20 qubits both are worse, register 8.87% and matched MLP 8.71% (seed 1). The per-variant counts in Table 17 are not independent evidence, because all 16 variants are scored by one trained policy and move together from run to run. The early lead does not survive full training, and, as on the physics rows, a matched classical core does what the register does. With the 3-D tethered register at 30 qubits (EAL protocol, common random numbers, bf16 mixed precision, seed 1, 10 epochs), the register's gap is 8.62% against 8.77% for its matched MLP (1,715,198 and 1,715,057 parameters) and 8.69% for vanilla at seed 1; over seeds 1–3 the register, the MLP and vanilla average 8.67%, 8.68% and 8.63% (vanilla seeds 2–3 run on a second machine whose seed-1 rerun lands within 0.03 points). Against the published EAL result (8.58%, the unweighted mean of Table 11 in Berto et al. (2025)), fp32 vanilla runs give 8.53, 8.60 and 8.82% (seeds 1–3, mean 8.65%: parity, not a lead); the first was 0.05 points lower and the bf16 runs are 0.04–0.19 points higher; on 2,048 generated validation instances, decoding in fp32 instead of bf16 changes route cost by $+0.001\%$ (CI $\pm 0.05\%$), so the difference sits in training, not decoding. With the routed register (30 atoms, the design of Table 1, top) over five seeds in bf16, the mean gap is 8.61% against 8.65% for vanilla (better on 2 of 5 seeds; Table 1); with the sampling decode below it is 5.96% against 6.02% (3 of 5). Neither difference is outside the seed spread (standard deviation of the paired difference 0.19 points). A registered gate that applies the register only to time-window instances gives 8.73%.

Extra decode compute (registered, test files, reported apart). Multi-start sampling (temperature 0.5, four samples per start, eight augmentations; $4\times$ the protocol's decode) lowers the register's mean gap from 8.62% to 6.03% and the matched MLP's from 8.77% to 6.14% (a harness rerun of

the protocol decode reproduces every per-variant gap within 0.008 points). Register minus MLP is -0.11 points (95% CI -0.13 to -0.09 over test instances, 8 of 16 variants; one trained policy per arm, so the interval does not cover seed variation). These rows use more test-time compute than the published protocol and are not compared with 8.58% as like for like. Post hoc, at seed 1 the register was ahead on all eight time-window variants and behind on the other eight; a registered test on seeds 2 and 3 did not replicate this (register minus MLP: -0.15, $+0.13$, 0.00 points at seeds 1–3).

Blockade move selection (registered screen, VALID only). After the fine-tuned policy builds a route, ten rounds of local search take the 18 best improving 2-opt and relocate moves, keep a set that do not conflict, and apply them together. An ideal blockade array (adiabatic sweep, 100 shots) picks that set as a maximum-weight independent set; exact enumeration is the ceiling. On 128 VALID instances of four unseen variants and two seeds, applying compatible moves together lowers the mean gap from 14.55% (policy alone) to 8.39% (exact), against 9.43% for one move per round. On average the array was level with exact (8.38% against 8.39%, descriptive; no cell significant after Holm correction), but greedy selection was only 0.014 points above exact (8.40%), and from a nearest-neighbour start the array fell up to 0.15 points behind exact. A control that samples near-optimal sets at random matched the array on 2 of 4 variants, short of the registered 3 of 4 for attributing an edge to random selection; this negative control does not identify a role for the blockade dynamics either. The registered promotion to TEST required the array to reach the ceiling in every cell; one policy-start cell missed by 0.026 points, so no TEST run was made. Exact enumeration of $n \leq 18$ moves is instant, so this checks the emulation, not a speed-up. The one robust result is that, at equal rounds in this fixed pipeline, applying compatible moves together beats one move per round: a method-level effect, not quantum evidence. At these sizes the choice of selector had little observed effect.

Table 17: RouteFinder EAL, 16 multi-depot variants: mean test gap to the reference solutions (lower is better). Every arm trains on the same generated instances. Pilot: 1 epoch of 2,000 instances.

Budget	arm	seeds	mean gap (%)	variants better than vanilla
pilot	vanilla (K=0)	1	24.48	—
	register, K=5	1	23.66	14 of 16
	matched MLP, K=5	1	23.80	15 of 16
	register, K=10	1	24.31	11 of 16
	matched MLP, K=10	1	24.98	2 of 16
	register, K=20	1	24.36	11 of 16
	matched MLP, K=20	1	23.61	14 of 16
	register, K=30	1	23.91	13 of 16
	matched MLP, K=30	1	25.05	4 of 16
EAL protocol	vanilla (K=0)	1	8.53	—
	register, K=20	1	8.87	0 of 16
	matched MLP, K=20	1	8.71	3 of 16
	register, K=30	1	8.59	6 of 16
	matched MLP, K=30	1	8.56	8 of 16

S FUTURE WORK: FOUNDATION VIDEO WORLD MODELS AND LONG-HORIZON GENERATIVE METRICS

While the paper's title and framework refer to *physics world models*, our empirical results in this work focus on continuous neural PDE surrogates (The Well, RealPDEBench) and physical transport simulators (TokaMark, TORAX), with one discrete routing row (Appendix R). We explicitly leave the evaluation of open-domain, generative video world models and their long-horizon benchmarks as future work.

Video world model benchmark suites. Emerging benchmarks evaluate whether generative video models capture intuitive physical laws from visual inputs. Specifically, Li et al. (2025) introduce *WorldModelBench*, using visual question-answering judges across physical scenarios, while Punzo et al. (2026) develop *Physics-IQ* to score video prediction models against counterfactual physical

violations. Open-weights video foundation models such as Wan2.2 (Team Wan et al., 2025) and LingBot-World (Gao et al., 2026), and the 7B world-action model FLUX 3 Action (Black Forest Labs, 2026), are multi-billion-parameter diffusion or flow-matching models. FLUX 3 Action decodes future video frames and robot actions from the same future tokens; that shared latent is where the co-processor would be inserted, which we have not tested. On our hardware, one Physics-IQ pass costs 13–27 GPU-hours for three arms, and WorldModelBench needs a VLM judge pipeline on top. We make no empirical claims on video world model benchmarks in this paper; doing so under matched classical controls is an active direction for future work.

Long-horizon rollouts. The one candidate regime, rollout-aware fine-tuning evaluated on long-horizon rollouts (§5.2), did not replicate at four seeds; more seeds and a checkpoint rule that early checkpoints cannot dominate are the first requirement. Generative video world models drift away from physically consistent states over multi-second rollouts. Testing a routed register in the latent bottleneck of such a model, under the same matched controls, is future work.

Predicting the qubit scaling law. Appendix P measures the hybrid's error against the qubit budget and bounds it by a data-derived coverage curve; how efficiently qubits fill that curve is measured, not predicted. Recent theory could predict it. For an analog register the accessible frequencies are gaps of its Hamiltonian's spectrum, so per-input atom positions can give input-conditioned frequency support (Wang et al., 2026b) (in our registered test, swapping positions between inputs or fixing them to a grid changed VALID error by at most 0.21%, so this support went unused; Appendix P); Fourier-coefficient variance decays with frequency under non-linear embeddings such as our $\tanh$ detunings, a smooth bias consistent with our probe (Kang et al., 2026); pulse-level control changes the optimization landscape more than the reachable function class (Strobl et al., 2026); and finite-shot generalization bounds fix how a readout budget should be split between inputs and shots (Catak, 2026). Raising the coverage ceiling needs a richer or input-adaptive mode basis rather than more qubits.

Long-horizon metrics. One-step scores reward the behaviour we measure in the incoherent bands (Appendix I): where the prediction's coherence with the true change is low, the least-squares answer shrinks that band, and over a rollout the shrinkage compounds into lost small-scale energy. Long-horizon evaluation of world models should therefore report, per Fourier octave and horizon, the prediction-to-truth energy ratio and the coherence, the lag-one regression slope of each band's change (truth against model), and a proper probabilistic score (CRPS) for ensembles, next to each benchmark's own rollout metric. We have not run these; they are future work.

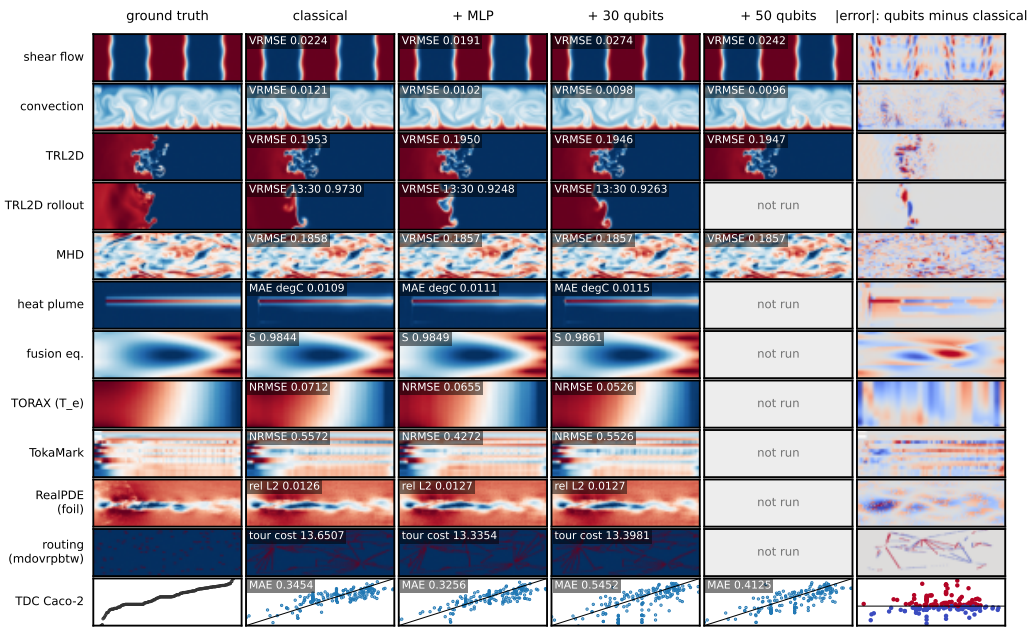


Figure 17: Figure 2 at full size, with notes per row (The Well rows show the first output channel; their printed VRMSE scores all channels of the sample); last column: change in absolute error from classical to qubits (blue: qubits better). Heat plume: the three predictions agree; the qubits' extra error (MAE 0.0115 against 0.0109 °C) sits on the steepest tenth of the plume (91% of the net excess), and elsewhere the net error change is small. Fusion equilibrium (PCA backbone, held-out TEST; panels give the split's mean official score S): classical 0.9844, MLP 0.9849, qubits 0.9861; the flux maps agree to R^2 0.9993, and the qubits' gain is in the shape of the flux surfaces, not visible at this scale. TORAX (vertical axis: time along a 32-step rollout; horizontal: normalized radius): on this sample (seed 1337) the clearest gain (0.0526 against 0.0712; MLP 0.0655), growing with lead time and largest at the magnetic axis; over three seeds the register and its MLP are close (VRMSE@8 0.0305 against 0.0309, classical 0.0321; register ahead on 1 of 3). RealPDEBench: the MLP and register corrections differ by less than 1% of the field's spread and leave the U-Net's wake unchanged. TRL2D rollout (VALID-selected rollout checkpoints, seed 1337; official start; the trajectory with the median classical error; density at step 22 of 30): on this trajectory the MLP and the register stay closer to the truth than the classical model (13:30 VRMSE 0.925, 0.926 and 0.973); over the nine test trajectories at this seed the register is lowest (0.993 against 1.033 MLP and 1.079 classical), an edge that does not replicate over four seeds (Table 12). Shear flow, convection and TRL2D (routed co-processor of Table 1, top, seed 1337; window fixed in advance): on the shear sample the MLP is lowest (VRMSE 0.0191; 0.0274 at 30 qubits, 0.0242 at 50, classical 0.0224), although over the test split 30 qubits tie the MLP (0.0213) and 50 qubits are lower (0.0210; classical 0.0237); on convection the sample follows the split (0.0096 at 50 qubits, 0.0098 at 30, 0.0102 MLP, 0.0121 classical; test 0.0425, 0.0433, 0.0435, 0.0458). On TRL2D (window 436) the arms differ only along the front (0.1946 at 30 qubits, 0.1947 at 50, 0.1950 MLP, 0.1953 classical; test 0.1921–0.1923). Routing (RouteFinder, first test instance of the first unseen multi-depot variant, MD-OVRPBTW; no reference tour is released, so the first column shows depots and customers, and the last column only marks where the register's and the classical tours differ, not an error): on this instance the MLP's tour is cheapest (13.34 against 13.40 for the register and 13.65 classical); in this 3-D run the register's mean gap over the 16 variants is lowest (8.62% against 8.69% classical and 8.77% MLP at seed 1; over three seeds 8.67% against 8.68% MLP and 8.63% classical), while at the EAL budget of Table 17 no routed arm beats vanilla (Appendix R). TokaMark (FT-Base recipe; the middle test shot by id, chosen mechanically after the scores were known, 467 windows left to right; rows: the 15 EFIT signals, two for each X-point; blank: no EFIT value): on this shot the MLP is lowest (NRMSE 0.43 against 0.55 register and 0.56 classical); over all 1,020 test shots the task score is 0.1500 MLP, 0.1504 classical and 0.1512 register (one seed, descriptive; our best, 0.1379, uses the Scratch recipe). MHD (routed, test window 480, mid-plane slice): the four arms agree to 0.0001 (VRMSE 0.1857–0.1858; test 0.1733 each), so the gate carries the whole gain.

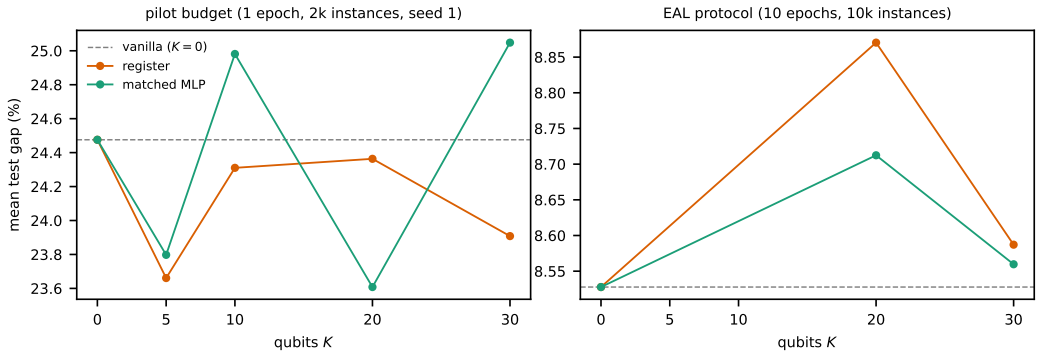


Figure 18: RouteFinder qubit ladder with the pretrained model fixed as the classical part ($K = 0$, dashed). Left: pilot budget. Right: EAL protocol. Dots are seeds.